

On the Underlying Long Vowels in Contemporary Standard European Portuguese

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Abstract:

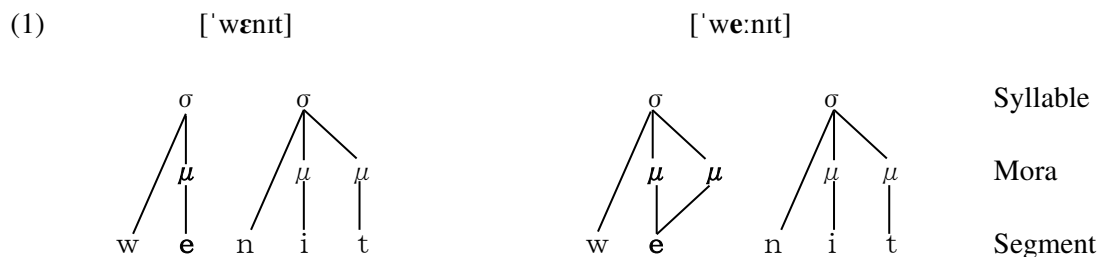
Traditionally, Portuguese linguistics has acknowledged phonetic differences in vowel length conditioned by stress in Contemporary Standard European Portuguese (EP), yet has rarely addressed the possibility that underlying vowel length oppositions — not directly inherited from but analogous to those attested in Classical Latin and established through phonological developments in the 14th–15th centuries — may persist in the Contemporary language. In contrast, Makino (2006¹, 2006², 2007²) argued for the persistence of an underlying opposition in vowel length even in Contemporary EP, seeking to account for a range of phonological phenomena: the characteristic vowel alternations in open syllables preceding prepalatal and palatal consonants [ʃ ʒ ʎ ɲ]; the phonetic outcomes of vowel coalescence at the level of phrasal phonology; vowel opening in radical vowels of derived nouns and adjectives formed by suffixation to verbal radicals lacking intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative implicatures. This theoretical study revisits this issue and offers a novel account of two phenomena: the absence of vowel quality reduction in nasal vowels in unstressed syllables, and the opposition [ɛ:] ≈ [á:] in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives. The analysis is developed within the framework of Stratal Optimality Theory (Stratal OT), and employs a revised view of vowel feature geometry in which the traditional [±ATR] feature is reinterpreted as the terminal direction of a multistage tongue-height vector. Without introducing new phonetic data — except for a key pair of instrumental measurements that provide acoustic evidence for the underlying length distinction — this study offers a principled reinterpretation of phonological patterns that have previously been considered beyond the scope of systematic analysis.

1. Introduction

In Contemporary Standard European Portuguese (henceforth EP), phonetic alternations in vowel length conditioned by stress can be observed — for example, [é:]ma "he/she loves" versus [ɐ]mamos "we love". However, as far as we are aware, little to no attention has been paid to the possibility that EP, like Classical Latin, may exhibit a phonological — that is, underlying — opposition in vowel length, as in *vĕnit* "he/she comes" versus *vēnit* "he/she came". When one considers not only the stress-conditioned alternations in vowel length, but also the presence or absence of vowel quality alternations under the same conditioning, along with the etymological backgrounds of the relevant forms, it becomes plausible to hypothesise that a phonological opposition between long and short vowels — established around the year 1500 — may still persist at the underlying level in contemporary EP. Moreover, positing such an opposition makes it possible to offer a more unified analysis of phonetic patterns that might otherwise appear unrelated. Accordingly, this study does not aim to introduce new phonetic data, but rather constitutes a theoretical reinterpretation of well-known facts from an entirely different perspective.

2. Phonological Vowel Length

In this section, we adopt the view that distinguishes between the segmental tier — which encodes phonemic quality — and the moraic tier — which encodes phonological quantity — in representing vowel length. Classical Latin provides a canonical case of an underlying vowel length opposition, as in the minimal pair *vĕnit* ['vɛnit] "he/she comes" vs. *vēnit* ['ve:nit] "he/she came". Their respective phonological representations may be schematised as in (1). (In the representations below, the final consonant /t/ is hypothetically treated as moraic; however, since the present analysis is concerned primarily with vowel quantity, the moraicity of /t/ is not crucial to the argument.)



The two forms are identical in that the segmental tier consists of the same sequence of segments

— /w/, /e/, /n/, /i/, /t/ — in that order. The difference lies in whether the vowel segment /e/ is linked to a single mora (μ) or to two moras. A mora is a unit of phonological weight and serves as a constituent of the syllable (σ) determining syllable weight — light vs. heavy. From the perspective of segment-to-mora association, a vowel segment (V) linked to a single μ yields a short vowel — $\overset{\mu}{/V/}$ (e.g. $\overset{\mu}{/wenit/}$ [$'wenit$]) — whereas one linked to two moras corresponds to a long vowel — $\overset{\mu\mu}{/V/}$ (e.g. $\overset{\mu\mu}{/wenit/}$ [$'we:nit$]). As shown in (1), onset consonants are not associated with any mora and therefore do not contribute to phonological weight.

3. Underlying moraic specification of vowel segments in Contemporary EP

In this section, we examine how vowel segments in Contemporary EP are associated with moras, from the theoretical perspective outlined in Section 2. A key piece of evidence for this investigation is the pattern of phonological alternation in 3.1.

3.1. Vowel Alternations Conditioned by Stress

In Contemporary EP, vowels typically exhibit phonetic alternations in both **quality** and **quantity**, as illustrated in (2), depending on whether they occur in stressed or unstressed syllable — for simplicity, the discussion is for now restricted to open syllables — within the same morpheme. Teyssier (1984: 20) refers specifically to the alternation in vowel quality as "réduction". In certain morphemes, by contrast, only alternations in **quantity** are observed, without accompanying changes in quality, as shown in (3)¹. (Note: the symbol $\approx$ indicates alternation, m.pt. stands for Middle Portuguese, < denotes a sound change, and $\square$ marks a borrowing.)

(2)	$[\acute{e}:] \approx [\textbf{i}]$	$pr[\acute{e}:]ga \approx pr[\textbf{i}]gar$ (< m.pt. <i>pregar</i> < lat. <i>plicāre</i>) from <i>pregar</i> "to nail"
	$[\acute{a}:] \approx [\textbf{e}]$	$[\acute{a}:]ta \approx [\textbf{e}]tar$ (< m.pt. <i>atar</i> < lat. <i>aptāre</i>) from <i>atar</i> "to tie"
	$[\acute{o}:] \approx [\textbf{u}]$	$m[\acute{o}:]ra \approx m[\textbf{u}]rar$ (< m.pt. <i>morar</i> < lat. <i>morāri</i>) from <i>morar</i> "to live"

¹ Makino (2006³) demonstrated acoustically that there is a statistically significant durational difference between stressed and unstressed syllables: the former are realized, on average, with approximately twice the duration of the latter. Since no substantial durational difference was observed in the consonantal segments, this difference in syllable duration is attributable primarily to vowel length.

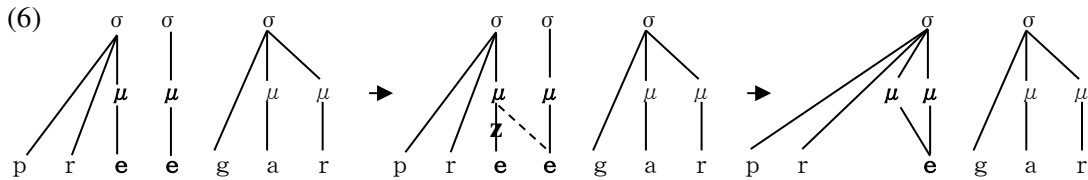
(3)	[ɛ:] ≈ [ɛ]	<i>pr[ɛ:]ga</i> ≈ <i>pr[ɛ]gar</i> (< m.pt. <i>preegar</i> < lat. <i>praedicāre</i>) from <i>pregar</i> "to preach" <i>refl[ɛ:]te</i> ≈ <i>refl[ɛ]tir</i> (< m.pt. <i>reflectir</i> □ lat. <i>reflectire</i>) from <i>refletir</i> "to reflect"
	[á:] ≈ [a]	<i>g[á:]nha</i> ≈ <i>g[a]nhar</i> (< m.pt. <i>gaanhar</i> < lat. <i>*guanjanjāre</i>) from <i>ganhar</i> "to gain" <i>[á:]to</i> ≈ <i>[a]tor</i> (< m.pt. <i>actor</i> □ lat. <i>actōre</i>) from <i>ato</i> "act" and <i>ator</i> "actor"
	[ɔ:] ≈ [ɔ]	<i>c[ɔ:]ra</i> ≈ <i>c[ɔ]rar</i> (< m.pt. <i>coorar</i> < lat. <i>colorāre</i>) from <i>corar</i> "to brush" <i>ad[ɔ:]ta</i> ≈ <i>ad[ɔ]tar</i> (< m.pt. <i>adoptar</i> □ lat. <i>adoptāre</i>) from <i>adotar</i> "to adopt"

From an etymological perspective, the vowel involved in the alternation in (2) derives from a monophthong (e.g. *prega* ≈ *pregar*), whereas the vowel in (3) originates in many cases from one of the following sources (4) and (5):

(4) **a sequence of two monophthongs** — resulting in a long vowel via contraction (e.g. *preega* ≈ *preegar*);

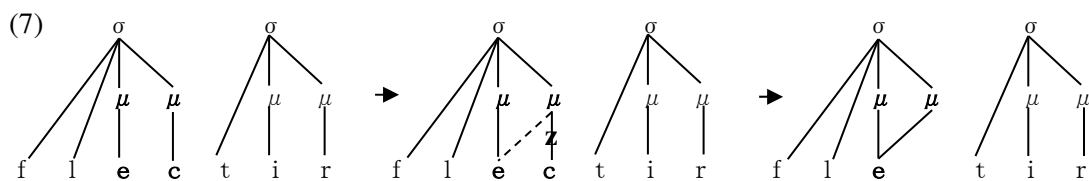
(5) **a monophthong followed by a coda consonant** — yielding a long vowel through compensatory lengthening triggered by consonant deletion (e.g. *reflecte* ≈ *reflectir*).

Assuming the two-tiered representation proposed in Section 2 — which distinguishes between the segmental tier and the moraic tier — the historical changes that the m.pt. *preegar* is likely to have undergone by "por volta de 1500" (cf. Teyssier 1980: 43) may be schematically represented as in (6):



In the initial stage of (6), the two adjacent pretonic vowels *e* were each linked to a single mora, forming two distinct syllables: *pre*^μ- and *-e*^μ. However, as a result of *contracção* — a vowel coalescence process that occurred during the 14th to 15th centuries (cf. Teyssier 1980: 40-43), likely triggered by the Obligatory Contour Principle (OCP) — the first *e* was deleted. Its associated mora was then linked to the following *e*^μ, resulting in compensatory lengthening. Simultaneously, resyllabification took place, yielding a new syllable *pre*^{μμ}- with an underlying long vowel /*ε*/ as its nucleus.

In the case of *reflectir* in (3), a borrowing from Classical Latin, the historical development follows essentially the same pattern, as illustrated in (7). For reasons of space, the prefix *re-* is omitted in the representation.



In this case, it is assumed that the coda consonant *-c* was deleted due to the tendency of PE to avoid closed syllables (cf. Makino 2004²: 207) — indicated by the delinking of *-c*, represented by **z** on the association line — and that the mora previously linked to *-c* was associated to the preceding ^μ*e*, resulting in a underlyingly long vowel /^{μμ}ε/ linked to two moras.

As Teyssier (1980: 42) notes: "No século XV, quando as contracções das vogais em hiato se completaram, essas vogais deviam ser *longas e abertas*, em oposição às pretónicas simples [e] [ä] e [o], que eram *breves e fechadas*", which would be in IPA [e] [ɐ] and [o], respectively. Although the open-mid long vowel /^{μμ}ε/ subsequently lost its phonetic length, it has retained its vowel quality to this day, as seen in *pr[ε]gar* and *refl[ε]tir*. On the other hand, the pretonic *e* in ^μ*pregar* in (2), which was etymologically a monophthong, did not undergo such historical changes and is assumed to have remained an underlying short vowel /^με/ linked to only one mora. After passing through the stage of [e] "*breve e fechada*" above in the 15th century, it subsequently underwent a phonetic change "no século XVIII, provavelmente depois de 1750" (Teyssier 1980: 62), and is today realised in the form *pr[i]gar*.

This situation is assumed to hold not only for *e* /ε/, but also for *a* /a/ and *o* /ɔ/, and likewise for stressed vowels. Accordingly, between the alternations in vowel quality and quantity conditioned by the presence or absence of stress in the Contemporary language and their earlier states, a relationship of the type illustrated in (8) can be observed:

(8)

Middle EP (ca. 1500)		Contemporary EP
Underlying short vowels: / ^μ é á ó/ ≈ / ^μ ε a ɔ/	⇒	[é: á: ó:] ≈ [i ɐ u]
Underlying long vowels: / ^{μμ} é á ó/ ≈ / ^{μμ} ε a ɔ/		[é: á: ó:] ≈ [ε a ɔ]

3.2. The Opposition between Short and Long Vowels at the Underlying Level and its Phonetic Realisation

What becomes clear from the vowel alternations in Contemporary EP illustrated in (8) is the following:

(9) The alternation in quantity — [é: á: ó:] ≈ [i ɐ u] and [é: á: ó:] ≈ [ɛ a ɔ] — is determined by the presence or absence of stress: vowels are realised as bimoraic long vowels in stressed syllables — especially in stressed open syllables — and as monomoraic short vowels in unstressed syllables.

(10) The alternation in quality — [é: á: ó:] ≈ [i ɐ u] — is likewise governed by stress, with vowel quality *réduction* occurring in unstressed syllables.

(11) Only three cases in which vowel quality *réduction* does not occur — such as those exemplified in (3) — are not synchronically predictable.

The alternation in quantity (9) and the alternation in quality (10) are both predictable from the presence or absence of stress and are therefore non-distinctive. By contrast, (11) is unpredictable and constitutes lexical information. Although previous studies in generative tradition have not made this point explicit, it may be inferred that both the vowels showing qualitative alternation — [é: á: ó:] ≈ [i ɐ u] — and those that do not — [é: á: ó:] ≈ [ɛ a ɔ] — are underlyingly represented by the same vowel segments /ɛ a ɔ/, and that the absence of *réduction* in vowel quality is thus considered a lexical idiosyncrasy specific to the lexical item in question.

In contrast, the present study proposes the hypothesis that **the underlying opposition between short and long vowels, established by the year 1500, has persisted into Contemporary EP.** According to this hypothesis, **the qualitative alternation pattern [é: á: ó:] ≈ [i ɐ u], which exhibits vowel quality reduction, corresponds to the phonetic realisation of underlying short vowels / $\overset{\mu}{\epsilon} \overset{\mu}{a} \overset{\mu}{\omicron}/$, whereas the absence of such alternation — as in [é: á: ó:] ≈ [ɛ a ɔ] — reflects the phonetic realisation of underlying long vowels / $\overset{\mu\mu}{\epsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\omicron}/$.**

Under this interpretation, **both underlying short vowels and underlying long vowels undergo some form of reduction in unstressed syllables.** Specifically, **the underlying long vowels**, which consist of two moras, **weaken by losing one mora** — that is, by reducing their length — while **the underlying short vowels**, which contain only one mora and thus have no mora to lose, **instead undergo vowel quality reduction as a means of weakening.** Moreover, the presence or absence of vowel quality reduction in (10) also becomes predictable as a reflex of the opposition between underlying short vowels / $\overset{\mu}{V}/$ and underlying long vowels / $\overset{\mu\mu}{V}/$. That is, in unstressed syllables, the short vowels / $\overset{\mu}{\epsilon} \overset{\mu}{a} \overset{\mu}{\omicron}/$ — linked to a single mora in the underlying representation — undergo vowel quality reduction and are phonetically realised as [i ɐ u], whereas the long vowels / $\overset{\mu\mu}{\epsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\omicron}/$ — linked to two moras in the underlying representation — retain their quality and are phonetically realised as [ɛ a ɔ]. Therefore, **the only information that**

speakers must store at the lexical level is whether or not a given lexical item contains an underlying long vowel.

To summarise the above discussion: the phonological opposition in vowel quantity that once existed — as shown on the left-hand side of (8) — is still preserved in the underlying representation today, as an unpredictable and oppositive property; in contrast, the vowel alternations shown on the right-hand side of (8), both in length and in quality, are predictable from the presence or absence of stress.

A key clue in support of this hypothesis lies in the occasional occurrence of longer phonetic realisations of the vowels involved in the alternation [é: á: ó:] ≈ [ε a ɔ], even in phonologically unstressed syllables. For instance, the vowel in *p[á:]da* (< lat. **pānāta*-) "small bread made with poor-quality flour" and its stress-shift counterpart *p[a]deiro* (< Lat. **pānatāriu*-) "(male) baker" and its derived form *p[a]deira* "(female) baker" is often realised as [a:] in *p[a:]deiro* and *p[a:]deira*, in contrast to the always short and close [ɐ] in *m[ɐ]deira* (< lat. *māteriam*) "wood (material)"². In my personal observation, unstressed syllables containing the non-reduced vowel [a] — particularly when they function as the head of a metrical foot, as in *p[a:]daria* (← *pada* + *-aria*) "bakery" — are often realised with a noticeably longer duration, as in *p[a:]daria*. This contrasts with cases like the personal name *C[ɐ]tarina* "a female given name, equivalent to English *Catherine*", where the foot head contains the reduced vowel [ɐ], which is clearly shorter.

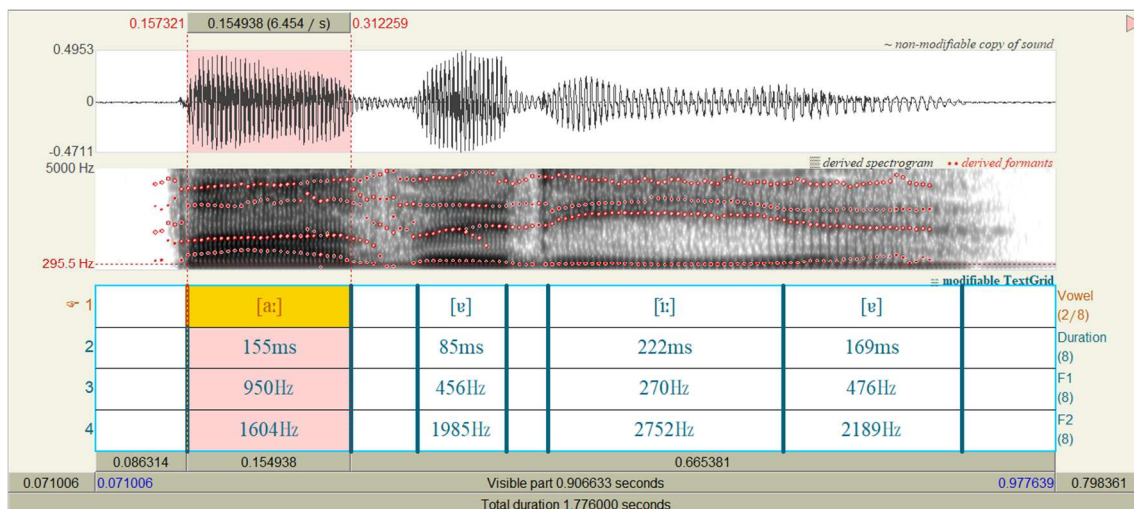
The following are spectrographic analysis images of the audio files available on *Infopédia*, referenced in footnote 2. Figure (12) corresponds to *padaria*, and Figure (13) to *Catarina*. The analysis was conducted using Praat (version 64.1.7). Although no information is provided regarding the speaker, it is a native female speaker, and the same person for both recordings. The overall word durations are 743 ms for *p[a:]daria* and 724 ms for *C[ɐ]tarina*, and are thus nearly identical. Both words also share the same metrical structure — namely, (*p[a:]da*)(*'ria*) and (*C[ɐ]ta*)(*'rina*) — and the two vowels under comparison both constitute the head of the secondary-stressed foot, not the primary-stressed position. In addition, the vowels that follow them are entirely identical: namely, [-ɐ-], [-i:-], and [-ɐ].

² As of 5 June 2025, the pronunciations of *pada*, *padeiro*, *padeira*, *madeira*, *padaria*, and *Catarina* can be accessed via the *Infopédia* online dictionary at the following entries:

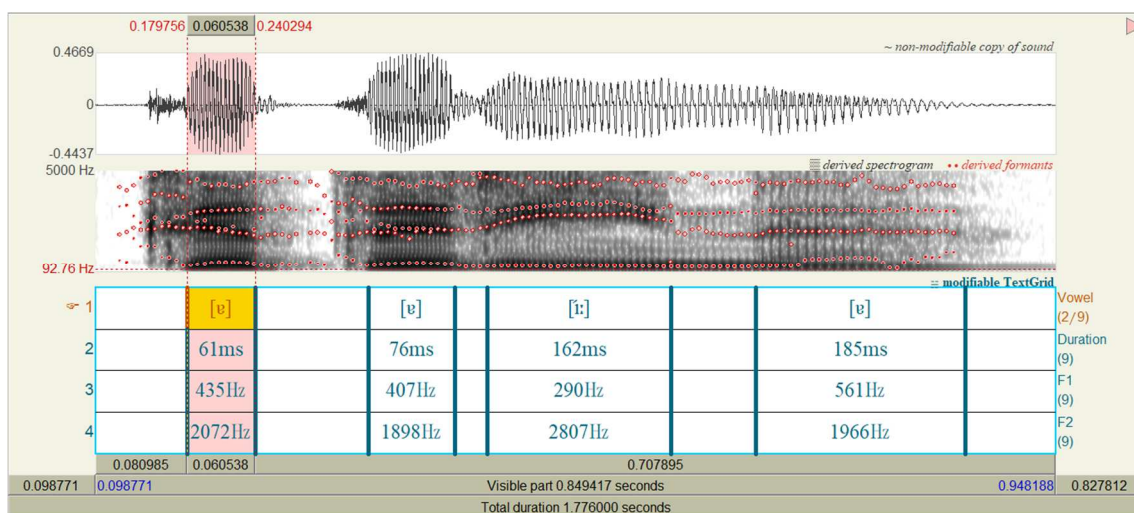
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/pada>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeiro>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeira>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/madeira>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padaria>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/catarina>

The vowel [a:] in *p[a:]daria* is particularly long and warrants close auditory attention.

(12) *padaria* [pa:ðə'ri:v]



(13) *Catarina* [kətə'ri:nə]



The acoustic measurements of the foot *pada-* in the noun *padaria* and of the foot *Cata-* in the noun *Catarina* are given in (14) and (15), respectively:

(14)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>pa-</i>	[a:]	155	950	1604
<i>-da-</i>	[v]	85	456	1985

(15)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>Ca-</i>	[v]	61	435	2072
<i>-ta-</i>	[v]	76	407	1898

In (14) the vowel [a:] in *p[a:]daria* exhibits markedly longer duration (155ms) and a more open

quality (F1 = 950Hz) than the corresponding vowel [ɐ] in *Catarina* (61ms, F1 = 435Hz) in (15). The striking contrast between *padaria* and *Catarina* is likely to provide definitive phonetic evidence that underlying bimoraic vowels /a/ are preserved in duration and quality in non-stressed foot-head positions in Contemporary Standard European Portuguese. It seems that this is not a mere hypothesis. This behaviour in terms of vowel length suggests that the vowels showing the alternation [é: á: ó:] ≈ [ɛ a ɔ] are underlyingly long vowels /^{μμ}ε ^{μμ}a ^{μμ}ɔ/, which have a strong tendency to be phonetically realised as short vowels under lack of stress, due to mora loss.

Another suggestive observation is that, in Contemporary EP, *length* serves as the phonetic implementation of stress (cf. Martins 1983). Accordingly, it is not surprising that underlyingly vowels /^μV/ tend to be phonetically lengthened when stressed, while underlyingly long vowels /^{μμ}V/ tend to be phonetically shortened when unstressed.

Finally, phonetic realisations of vowel coalescence at the phrasal level observed in Contemporary EP also provide strong evidence in support of this interpretation. For example, the final vowel -a in *fica* — the second person singular imperative form of the verb *ficar* "to stay" — and the initial vowel a- in the adverb *aqui* "here" are both unstressed, and when pronounced in isolation, each of these is typically realised as the reduced vowel [ɐ], as in *fic*[ɐ] and [ɐ]*qui*. When the two words are combined to form the imperative sentence *Fica aqui* "Stay here", however, it is far more natural in ordinary speech to pronounce them not as *Fic*[ɐ *ɐ*]*qui*, but rather as *Fic*[a]*qui*, with the non reduced open vowel [a]. This synchronic vowel crasis can be accounted for by the diagram in (16), assuming the hypothesis proposed above — namely, the opposition between underlying short vowels /^με ^μa ^μɔ/ and underlying long vowels /^{μμ}ε ^{μμ}a ^{μμ}ɔ/, and the phonological changes they undergo in unstressed syllables: /^με ^μa ^μɔ/ and /^{μμ}ε ^{μμ}a ^{μμ}ɔ/ → [i ɐ u] and [ɛ a ɔ]:

$$(16) \text{Fica}^{\mu\mu} \text{a}^{\mu} \text{qui} \rightarrow \text{Fica}^{\mu\mu} \text{a}^{\mu} \text{qui} \rightarrow \text{Fic}'^{\mu\mu} \text{a}^{\mu} \text{qui}$$

In more explicit terms, the final a of *fica*^μ and the initial a of *a*^μ*qui* are each underlying monomoraic vowels /^μa/. Therefore, when these words are pronounced in isolation, both vowels appear in unstressed position and undergo vowel quality reduction, being realised phonetically as [ɐ]. However, when the two words occur in sequence within a syntactic constituent, the adjacency of identical phonological elements is likely disfavoured by the OCP, leading to the deletion of the vowel quality of *fica*'s a, while its mora is retained and reassociated to the following a- of *a*^μ*qui*. As a result, a derived long vowel /^{μμ}a/ emerges, which is phonetically realised as a non-reduced open vowel [a]. Other examples of the same type of coalescence —

e.g. *dava-o* ['da:vɔ] "he/she/you used to give it" and *dava-a* ['dava] "he/she/you used to give it" shall be dealt with in Section 6.3.

4. From Underlying to Surface Representation

The next issue concerns how to account for the mapping from the underlying representations to their respective surface ones: $/\overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon}/ \rightarrow [\acute{\varepsilon}: \acute{a}: \acute{\upsilon}:] \approx [i \text{ } \text{v} \text{ } u]$ and $/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\upsilon}/ \rightarrow [\acute{\varepsilon}: \acute{a}: \acute{\upsilon}:] \approx [\varepsilon \text{ } a \text{ } \upsilon]$. In addition to $/\overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon}/$, the underlying inventory of short vowels in Contemporary EP also includes $/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon} \overset{\mu}{o} \overset{\mu}{u}/$. Taking these into account, it appears that in stressed open syllables — particularly before [–nasal] consonants — and in pretonic open syllables — especially when not in absolute word-initial position — the following complex mapping generally occurs: $/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon} \overset{\mu}{o} \overset{\mu}{u}/ \rightarrow [i: \acute{\varepsilon}: \acute{\varepsilon}: \acute{\upsilon}: \acute{\upsilon}: \acute{o} \acute{u}] \approx [i \text{ } i \text{ } i \text{ } \text{v} \text{ } u \text{ } u]$. This paper proposes that these mappings, including those described earlier, can be systematically derived within the framework of Stratal Optimality Theory (Stratal OT), as shown in (17):

(17)

$/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon} \overset{\mu}{o} \overset{\mu}{u}/$	$/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon} \overset{\mu}{o} \overset{\mu}{u}/$	$/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\upsilon}/$	$/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\upsilon}/$	Underlying representation
$\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\upsilon} \overset{\mu}{o} \overset{\mu}{u}$	—	$\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\upsilon}$	—	Stress assignment
—	$i \text{ } i \text{ } i \text{ } \text{v} \text{ } u \text{ } u$	—	—	Vowel quality modification
$i: \acute{\varepsilon}: \acute{\varepsilon}: \acute{a}: \acute{\upsilon}: \acute{o}: \acute{u}:$	—	—	$\varepsilon \text{ } a \text{ } \upsilon$	Vowel quantity adjustment
$[i: \acute{\varepsilon}: \acute{\varepsilon}: \acute{a}: \acute{\upsilon}: \acute{o}: \acute{u}:]$	$[i \text{ } i \text{ } i \text{ } \text{v} \text{ } u \text{ } u]$	$[\acute{\varepsilon}: \acute{a}: \acute{\upsilon}:]$	$[\varepsilon \text{ } a \text{ } \upsilon]$	Surface representation

Since the reduction of the vowel quality of underlying short vowels occurs in unstressed syllables, syllabification and stress assignment must already have taken place prior to that stage. Furthermore, since vowel quality reduction in unstressed syllables affects only underlying short vowels, the underlying distinction in vowel quantity must still be preserved at that stage. Under the current model, thus, the following strata and constraint rankings are being hypothesised:

(18) **Lexical Stratum 1: Feature Establishment**

(19) **Lexical Stratum 2: Syllabification**

(20) **Lexical Stratum 3: Stress Assignment**

(21) **Postlexical Stratum 1: Vocalic Quality Modification**

In this stratum, the following set of constraints is assumed to be interacting:

The inviolable constraint IDENT-QUANTITY;

IDENT(lowVhigh) >> *[+low]/V[–stressed, μ] >> *[–high]/V[–stressed, μ] >> IDENT(high);

*[+low]/V[–stressed, μ] >> IDENT(low);

*[±ATR]/V[–stressed, μ] >> IDENT(ATR);

*[–back]/V[–stressed, μ]__([–sonorant])# >> IDENT(high^back) >> *[–back]/V[–stressed, μ]
>> IDENT(back);

IDENT(high), IDENT(low), IDENT(ATR), IDENT(round) >> *[+high], *[–high], *[+low], etc.

(22) Postlexical Stratum 2: Vocalic Quantity Adjustment

In this stratum, the following set of constraints is assumed to be interacting:

MAX, DEP >> *μμμ >> *σ[μ] >> *μμ/V >> MAX(μ);

in parallel with

*σ[μ] >> *μμ/V, DEP(μ)

While Lexical Strata 2 (Syllabification) and 3 (Stress Assignment) are important background, they fall beyond the primary focus of this article. We shall therefore defer a full formalization of those levels to a subsequent paper. The present study shall concentrate exclusively on:

Postlexical Stratum 1: Vocalic Quality Modification;

Postlexical Stratum 2: Vocalic Quantity Adjustment.

4.1. Postlexical Stratum 1: Vocalic Quality Modification

In this stratum, the short vowels ^ui ^ue ^uε ^ua ^uɔ ^uo ^uu — hereafter simplified as i, e, ε, a, ɔ, o, u — undergo changes in vowel quality in unstressed syllables. The present study limits its discussion to those unstressed syllables that occur in word-medial open syllables and in word-medial closed syllables ending in /r/ or /s/³. By contrast, the long vowels ε: a: ɔ: pass through this

³ In medial unstressed syllables ending in /l/, the vowel reduction described in (9) does not occur (e.g. *feltar* "to felt" /feltrar/ → [fel'tra:r]; *selvagem* "wild" /selvazen/ → [sel'va:ʒẽ]; *faltar* "to lack" /faltar/ → [fal'ta:r]; *voltar* "to return" /vɔltar/ → [voɫ'ta:r]~[vɔɫ'ta:r]). In medial unstressed syllables closed by /n/, the vowel+/n/ sequence — which bears two moras — is assumed to undergo nasalisation in an earlier stratum — e.g. /in/ → ã: — and thus enters the current stratum as a bimoraic nasal vowel. Other environments — namely, absolute word-initial unstressed syllables, penultimate unstressed syllables,

stratum without undergoing any changes in either quality or quantity, and will instead be subject to quantitative adjustment in Stratum 4.2. Before turning to the feature changes affecting the short vowels in this stratum, however, it is necessary to first consider the nature of vowel aperture oppositions among the vocalic segments of Contemporary EP.

In representative generative phonological studies of Portuguese — namely, Andrade (1977), Mateus (1982), Mateus & Andrade (2000: 31–35), and Mateus et al. (2003: 998–1001) — an analysis such as (23) has been adopted⁴. In this framework, the feature [–low] is considered redundant for [+high] vowels, just as [–high] is considered redundant for [+low] vowels.

(23)

high				low
+	/i/	[i]	/u/	–
–	/e/	[e]	/o/	+
	/ɛ/	/a/	/ɔ/	

This interpretation of the vowel aperture appears extremely straightforward, but it presents several problems. First, if /ɛ a ɔ/ form a natural class on the basis of sharing the feature [+low] in terms of vowel aperture, then from the perspective of phonological theory, one would expect them to behave uniformly with respect to processes of vowel raising in unstressed syllables. However, in pretonic CV syllables, /ɛ/ and /ɔ/ are raised to the high vowels [i] and [u], respectively — that is, vowels specified as [+high, –low] — whereas /a/ is raised to the mid vowel [e], specified as [–high, –low]. If phonological processes are to be understood as being sensitive to distinctive features, this pattern is difficult to account for.

Second, in EP, /a/ undergoes raising to [é:] in stressed CV syllables preceding [m n ɲ], whereas /ɛ/ and /ɔ/ retain their underlying aperture. Within the aforementioned model, this leads to a further asymmetry: although /ɛ a ɔ/ are all analysed as [+low] and thus form a natural class, /a/ behaves differently from /ɛ/ and /ɔ/ with respect to raising. Consequently, if one assumes that /ɛ a ɔ/ form a natural class by virtue of their [+low] specification, one must then explain why /a/ raises while /ɛ/ and /ɔ/ do not, based on some additional feature that distinguishes /a/ from the

and final unstressed syllables — exhibit weakening patterns that vary depending on whether the syllable is open or closed, and on the nature of the final consonant; due to the considerable complexity involved, these cases will not be addressed here.

⁴In Mateus & Andrade (2000), /i/ is treated as the least marked vowel in Portuguese and is subject to underspecification. As a result, /i/ and /u/ are not specified for either [high] or [low] in the underlying representation, and /e/ and /o/ are specified only as [–high], while /ɛ/, /a/, and /ɔ/ are specified only as [+low].

other two within the [+low] set. In their descriptive framework, /ɛ/ is distinguished from /a/ by its [−back] specification, and /ɔ/ is distinguished from /a/ by its [+round] specification. It would therefore follow that the features [−back] in /ɛ/ and [+round] in /ɔ/ somehow inhibit the [+low] → [−low] change from applying in stressed CV syllables preceding [m n ɲ]. But this seems to be an implausible explanation.

Third, in pretonic CV syllables in Brazilian Portuguese — henceforth PB — /a/ remains as the [+low] vowel [a], at least within the aforementioned framework, while /ɛ/ and /ɔ/ are raised to the [−low] mid vowels [e] and [o], respectively. This too is difficult to account for under the analysis that treats /ɛ a ɔ/ as forming a natural class of [+low] vowels. On the other hand, if /a/ is analysed as [+low] and /ɛ ɔ/ as [−low, −high, −ATR], then the three vowels do not constitute a natural class, and there is no expectation that they should pattern together phonologically in terms of aperture. Moreover, this raising can be analysed as involving a shift in feature specification: [+low] → [−low] in the case of /a/ → [e], and [−ATR] → [+ATR] in the case of /ɛ ɔ/ → [e o].

Fourth, in BP, furthermore, the sequences /-eia/ and /-eia/ are opposed even at the phonetic level, as seen in *feia* "ugly" /feia/ → [ˈfɛjɐ] and *ideia* "idea" /ideia/ → [iˈdɛjɐ], respectively. In EP, however, this opposition is neutralised: both /-eia/ and /-eia/ are realised as [−ɐjɐ], as in *feia* /feia/ → [ˈfɛjɐ] and *ideia* /ideia/ → [iˈðɛjɐ]. That is, [ɐ] functions as a central **mid** vowel that corresponds to both the front **mid-close** vowel /e/ and the front **mid-open** vowel /ɛ/, and as such, it contrasts with the central **open** vowel /a/, as in *faia* "beech" /faia/ → [ˈfajɐ]. The centralisation of /e/ and /ɛ/ before /i/ may be attributed to a restriction on the sequence of [−back] features, enforced by the OCP. The fact that both vowels are realised as [ɐ] and thus neutralised suggests that within the [+back, −round] region, the "aperture" [±ATR] opposition is not available as a oppositive feature in the system. If /ɛ/ were [+low] just like /a/, there would be no explanation for why *ideia* is not realised as *[iˈðajɐ].

From a theoretical perspective, it is reasonable to posit that the low vowel /a/ occupies the foundational position in the vowel system — articulatorily the most low and acoustically the most salient vowel. The high vowels /i/ and /u/ then arise as maximally opposed poles relative to /a/, completing a triangular system that ensures maximal perceptual opposition. Subsequently, the addition of /e/ and /o/ serves to fill the **mid**-space between these extremes, producing the commonly observed five-vowel system. Only later, as a form of system refinement, do languages introduce further oppositions between **mid-close** and **mid-open** vowels — /e/ vs. /ɛ/, and /o/ vs. /ɔ/ — distinctions that are structurally fragile and often subject to neutralisation as in

EP. This trajectory — from /a/ to /i u a/, then to /i u e o a/, and finally to /i u e o ε ɔ a/ — aligns well with both the articulatory-acoustic continuum and cross-linguistic patterns of vowel inventory complexity.

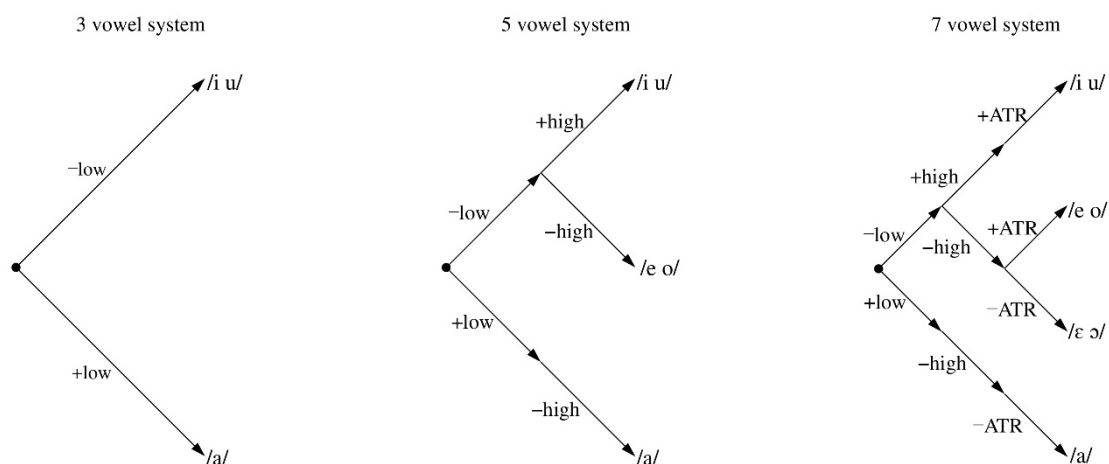
Based on the foregoing discussion, it seems more appropriate to analyse the vowel tongue height opposition among EP vocalic segments in terms of $[\pm\text{high}]$ and $[\pm\text{low}]$, as follows:

(24)

high				low
+	/i/	[i]	/u/	—
—	/e/	[ɐ]	/o/	
	/ɛ/		/ɔ/	
	/a/			+

The problem with this system (24) is that it lacks a feature capable of representing the opposition between /e o/ and /ɛ ɔ/, as indicated by the shaded cells. Given that the aforementioned studies consistently analyse this opposition as one between $[-\text{low}]$ and $[\text{+low}]$, and make no reference to $[\pm\text{ATR}]$ — and moreover, other researchers within the structuralist–functionalist tradition have likewise consistently treated this as an opposition of aperture — it appears that, for native-speaker researchers, this is self-evidently a tongue-height-based or aperture-based opposition. Accordingly, in this paper, the feature $[\pm\text{ATR}]$ — which is generally assumed to be dominated by the Pharyngeal Cavity node — is employed in italics as a convenient label for representing an additional tongue-height or aperture opposition within the $[-\text{high}, -\text{low}]$ region. This paper proposes the following systems of vowel tongue height:

(25)



In (25), [$\pm$ low], [$\pm$ high], and [$\pm$ ATR] are reinterpreted as features that encode the conceptual movement of the tongue for vowel segments within the vowel space — starting from a neutral position, represented by ● — toward their specific tongue height targets within the system: high /i, u/, mid-high /e, o/, mid-low / ϵ ɔ /, and low /a/.

First, [–low] and [+low] are taken to encode upward and downward movement, respectively, from the neutral position. Whereas in traditional systems [–low] expresses a neural command not to lower the tongue from the neutral position and, correlatively, the tongue position itself, in the present system it represents upward movement from that position. In a three-vowel system such as /i u a/, the endpoints of the [$\pm$ low] movement coincide with the upper and lower bounds of the vowel space, thereby determining the tongue height — or aperture — of each vowel: /i u/ are specified as [–low], while /a/ is specified as [+low].

By contrast, in a five-vowel system such as /i u e o a/, the feature [$\pm$ low] alone does not suffice to determine the tongue height of each vowel. From the endpoint of [$\pm$ low] movement, a subsequent [$\pm$ high] movement begins: [+high] indicates upward movement, and [–high] indicates downward movement. Thus, /i u/ reach their respective endpoints by undergoing a continuous upward movement specified first by [–low] and then by [+high], while /a/ reaches its endpoint through a continuous downward movement specified by [+low] followed by [–high]. In both cases, the endpoint of the movement coincides with either the upper or lower boundary of the vowel space, thereby terminating the motion. In contrast, the mid vowels /e o/, specified as [–low, –high], first rise under [–low], then change direction under [–high], resulting in an endpoint located at an intermediate tongue height between that of /i u/ and /a/. Note, however, that in actual articulation, the tongue does not first rise and then fall for /e o/; rather, they are produced directly at a mid-tongue position. The sequence [–low, –high] represents not a literal articulatory trajectory but rather the abstract vectorial structure by which each vowel segment's tongue height is determined. The markedness of mid vowels /e o/ in relation to both the high vowels /i u/ and the low vowel /a/ is thereby captured by the fact that the vector leading to their endpoint includes a point of inflexion — that is, a structural complexity.

Furthermore, in a seven-vowel system such as /i u e o ϵ ɔ a/, the features [$\pm$ low] and [$\pm$ high] alone are insufficient to determine the tongue height of each vowel. From the endpoint of the [$\pm$ high] movement, a further movement specified by [$\pm$ ATR] begins. [+ATR] represents an upward movement, while [–ATR] represents a downward movement. As already noted, it is important to bear in mind that in this paper, [ATR] in italics does not refer to tongue root advancement in the narrow sense. Accordingly, /i u/ undergo a continuous upward movement in

the sequence [–low, +high, +ATR], reaching their final endpoint; /a/, on the other hand, undergoes a continuous downward movement in the sequence [+low, –high, –ATR], likewise reaching its endpoint. In both cases, the endpoint coincides with the upper and lower bounds of the vowel space, and thus the movement terminates. The vowel /e o/, specified as [–low, –high, +ATR], initially ascends by [–low], then changes direction to descend via [–high], and then ascends again via [+ATR], ultimately reaching a mid-close tongue height as the endpoint. By contrast, /ε ɔ/, specified as [–low, –high, –ATR], first ascend via [–low], then descend via [–high], and then continue to descend via [–ATR], ultimately reaching a mid-open tongue height as the endpoint. In summary:

/a/ [+low, –high, –ATR]: **neutral position** → **descent** → **descent** → **descent**

/i u/ [–low, +high, +ATR]: **neutral position** → **ascent** → **ascent** → **ascent**

/ε ɔ/ ([–low, –high, –ATR]): **neutral position** → **ascent** → **descent** → **descent**

/e o/ ([–low, –high, +ATR]): **neutral position** → **ascent** → **descent** → **ascent**

To summarise the above, each of the features [±low], [±high], and [±ATR] corresponds to a single segment of the movement through which a vowel's tongue height — and consequently its aperture — is determined. Rather than functioning as coordinates that statically locate the vowel within the vowel space, these features behave as vectors that define the directional trajectory toward the vowel's articulatory endpoint.

What must be noted here is the asymmetry in markedness between the mid vowels /e o/ and /ε ɔ/ in the context of a seven-vowel system. While both sets of vowels are marked in that their tongue height is determined by a vector with an inflection point, /ε ɔ/ are less marked than /e o/ in that they involve only one inflection point, whereas /e o/ involve two. From this perspective, the vowels in a five-vowel system that correspond to /e o/ are not /e o/ themselves, but rather /ε ɔ/. In Contemporary EP, when scholarly words of Latin origin contain stressed vowels written as *e* or *o*, their quality is consistently [ε] and [ɔ] and not [e] and [o], regardless of vowel length in the Latin source. In other words, the opposition between /e o/ and /ε ɔ/ is neutralised in scholarly words. The fact that the outputs in such cases are [ε ɔ] — [–low, –high, –ATR] — and not [e o] — [–low, –high, +ATR] — may be attributed to the relative unmarkedness of [ε ɔ], whose defining vectors contain only a single inflection point.

Barbosa (1965: 76) notes that "/a/ est probablement légèrement plus fermé dans *sei* «je sais» que dans *cama* «lit»", and this, too, can be accounted for within the framework of the schema in

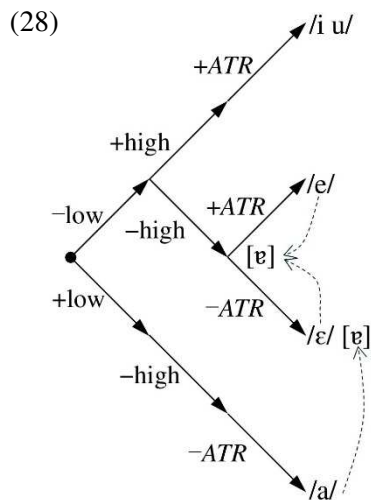
(25). The vowel /a/ — namely [ɐ], as he considers the stressed -e- in *sei* and the stressed -a- in *cama* to be phonetic realisations of the same phoneme /a/ — can be understood, in the case of *s[ɛ̃]i*, as an allophone that results from the following process:

(26) The feature [-back], shared by /e/ (specified as [-low, -high, +ATR, -back, -round]) and /ɛ/ ([-low, -high, -ATR, -back, -round]), is prohibited by the Obligatory Contour Principle (OCP) from occurring adjacent to the [-back] feature of a following /i/. As a result, /e/ and /ɛ/ undergo a change from [-back] to [+back].

(27) Through the shift [-back] → [+back], /e/ and /ɛ/ move into the central mid vowel region, defined by the feature set [-low, -high, +back, -round]. In this region, the distinctive opposition in [±ATR] becomes impossible to maintain, leading to the deletion of the [±ATR] specification from both segments. As a result, they are both realised as [ɐ], a vowel lacking any ATR specification and characterised by **[-low, -high, +back, -round]**.

The forms discussed above — *feia* /feia/ → [ˈfɛ̃ɐ] and *ideia* /ideia/ → [iˈdɛ̃ɐ] — are precisely examples of this neutralisation of opposition.

On the other hand, the [ɐ] in *c[ɛ̃:]ma*, as examined in Section 6.4, is an allophone resulting from nasal resonance in the /a/, triggered by the spreading of the [+nasal] feature of [m], which diverts part of the pulmonic airstream into the nasal cavity. This reduces the amount of airflow available for oral articulation and renders the [+low] region unusable. Consequently, in the underlying form /a/ — specified as [+low, -low, -ATR, +back, -round] — only the feature [+low] is altered, changing to [-low], while all other features remain unaffected. Thus, the resulting [ɐ] in this context is specified as **[-low, -high, -ATR, +back, -round]**. That this [ɐ] and the one described above differ in tongue height is evident from (28):



As we shall see immediately below, the interpretation of tongue height or aperture degree in (25) also allows for a straightforward explanation of the tongue height involved in the reduction of /a/ to [ɐ] in unstressed syllables.

Returning to the feature changes affecting the short vowels in this stratum, they are illustrated in (26) below. The altered feature values are shaded for clarity:

(29)

	high	low	ATR	back	round
i	+	–	+	–	–
e	–	–	+	–	–
ɛ	–	–	–	–	–
a	–	+	–	+	–
ɔ	–	–	–	+	+
o	–	–	+	+	+
u	+	–	+	+	+

→

	high	low	ATR	back	round
i	+	–	0	–	–
ɪ	+	–	0	+	–
ɨ	+	–	0	+	–
ɐ	–	–	0	+	–
u	+	–	0	+	+
ʊ	+	–	0	+	+
ʊ	+	–	0	+	+

The underlying four degrees of tongue height — /i u/ vs. /e o/ vs. /ɛ ɔ/ vs. /a/ — are reduced to a two-way opposition in unstressed position: [i ɪ u] vs. [ɐ]. Since [+low] is no longer active in the system, and all vowels involved share [–low], the resulting opposition in tongue height is effectively reduced to a two-way contrast expressed solely by [±high]. Thus, the specification [±high] alone suffices, and [±ATR] becomes unnecessary and is left unspecified. In fact, the unstressed vowel [ɐ] differs in aperture from the stressed vowel [é], the latter of which is characterised as [–ATR]. Specifically, unstressed [ɐ] has an aperture closer to that of IPA [ə], a more closed central vowel. Emiliano (2009: 31–33) provides crucial observations on this point: while stressed [é] aligns in height with [ɛ] and [ɔ], unstressed [ɐ] is slightly higher and should be transcribed as [ə], positioned between [e o] and [ɛ ɔ] in terms of aperture. For example, the word *cama* should ideally be transcribed not as ['kəmə] but as ['kəməə].

In this weak position, vowel reduction is understood to proceed as follows:

(30) The centralisation of the low vowel /a/ occurs, resulting in /a/ → [ɐ], which corresponds to the feature change [+low] → [–low].

(31) As a consequence of (27), /e ɛ/, which share [–low, –high, –round] with the resulting [ɐ], undergo a feature change from [–high] to [+high] in order to preserve their underlying opposition with /a/, now realised as [ɐ].

(32) Furthermore, in order to maintain their underlying opposition to /i/, which surfaces as [i], /e ɛ/ also shift from [–back] to [+back], ultimately mapping /e ɛ/ to [ɪ].

(33) The feature [±ATR], which underlyingly distinguishes /e/ from /ɛ/, is based in EP on a fine-grained aperture opposition among segments sharing [–high]. Once /e ɛ/ become [+high], this opposition becomes untenable, and [±ATR] is accordingly neutralised: [±ATR] → [0ATR].

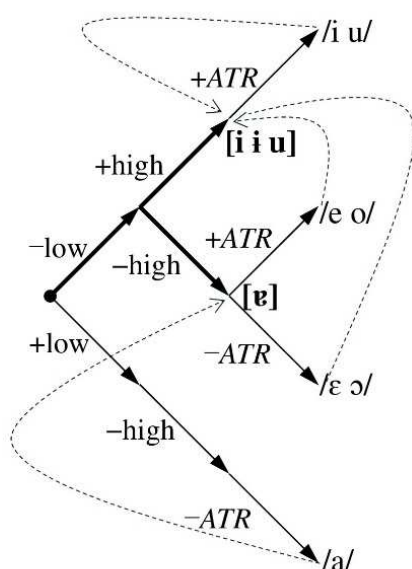
(34) The round back vowels /o ɔ/, which are specified as [–low, –high] and thus form a

natural class with /e ε/ in terms of tongue height, undergo parallel changes: [−high] → [+high] and [±ATR] → [0ATR], and are therefore mapped to [u].

(35) In order for /o ɔ/ now realised as [u] to preserve its opposition to the underlying /u/ realised also as [u], it would need furthermore t□ chang□ [+r□und] t□ [−r□und]; h□w□v□, doing so would yield [i], thereby neutralising the underlying opposition to /e ε/ realised as [i]. As a result, the system instead opts to neutralise the opposition between /o ɔ/ and /u/, maintaining [u] in both cases.

This vowel reduction can be schematised as in (33), where the dashed curve represents the process of vowel reduction:

(36)



As we shall see immediately below, the interpretation of tongue height or aperture degree in (25) also allows for a straightforward explanation of the tongue height involved in the reduction of /a/ to [e] in unstressed syllables. In terms of tongue height, this reduction is extremely simple: it consists merely in deleting all ATR values and raising the height by one step — [+low] becomes [−low] for /a/, and [−high] becomes [+high] for /e o ε ɔ/. For /e ε/, this change is accompanied by a shift from [−back] to [+back]. Accordingly, the vowel quality changes listed in (26) can be

classified into three distinct types. The first type concerns features related to vowel height:

(37) Raising of mid vowels: [−high] → [+high] (e ε o ɔ → i i u u)

(38) Raising of low vowels: [+low] → [−low] (a → e)

The second type involves the feature [ATR] and affects the entire vowel system:

(39) Non-utilisation of ATR within the system: [±ATR] → [0 ATR] (i e ε a ɔ u → i i i e u u u)

The third type pertains to the feature [back]:

(40) Centralisation of front mid vowels: [−high, −low, −back] → [+back] (e ε → i)

From (34) and (35), the descriptive generalisation (38) can be formulated:

(41) In unstressed syllables, non-high short vowels specified as [−high] and low short vowels specified as [+low] are prohibited. This requirement is enforced by the unfaithful mappings [−high] → [+high] and [+low] → [−low] (e.g. *bolacha* /bol-af-a/ → bu'lafɐ vs. *bo'l.afɐ; *casado* /kaz-ad-o/ → kɐ'zadu vs. *ka'zadu). However, the mappings [−high] → [+high] and [+low] → [−low] do not occur simultaneously; rather, only the latter takes place.

From this, the following constraint rankings can be inferred:

*[−high]/V[−stressed, μ] >> IDENT(high)

*[+low]/V[−stressed, μ] >> IDENT(low)

*[+low]/V[−stressed, μ] >> *[−high]/V[−stressed, μ]

The definitions of the above constraints are as follows:

IDENT(high): Assign one * for each change in the [high] feature between input and output.

IDENT(low): Assign one * for each change in the [low] feature between input and output.

*[−high]/V[−stressed, μ]: Assign one * for each unstressed non-high short vowel.

*[+low]/V[−stressed, μ]: Assign one * for each unstressed low short vowel.

This ranking alone cannot prevent, however, a candidate that simultaneously undergoes the mappings [−high] → [+high] and [+low] → [−low] — such as *casado* /kaz-ad-o/ → *ki'zadu — from incorrectly winning over the expected output kɐ'zadu. To resolve this, a higher-ranked faithfulness constraint must be introduced above the ranking *[+low]/V[−stressed, μ] >> *[−high]/V[−stressed, μ]. This constraint should penalise the simultaneous violations of both IDENT(low) and IDENT(high) — as in the mapping a → *i — while still favouring the mapping a → ɐ, which only violates IDENT(low). In doing so, the optimal candidate ɐ will correctly emerge as the winner over *i.

That is, the required constraint is one that allows for a change in either the [low] or [high] feature independently, but prohibits simultaneous changes in both. This constraint may be formalised as:

IDENT(low/high): Assign one violation mark if both [low] and [high] features change their values simultaneously.

This constraint is **indispensable for ensuring that /a/ preserves its underlying opposition with /e ε/, which surface as [i]**. From this, the following constraint rankings are inferred:

IDENT(lowVhigh) >> *[+low]/V[–stressed, μ] >> IDENT(low)

IDENT(lowVhigh) >> *[+low]/V[–stressed, μ] >> *[–high]/V[–stressed, μ] >> IDENT(high)

Of these, the ranking relationships between IDENT(low) and *[–high]/V[–stressed, μ], as well as between IDENT(low) and IDENT(high), remain undetermined. The reasons are as follows:

The winning candidates supported by IDENT(low) — namely, *i e ɔ o u* → *i i i u u u* — are all [+high], and thus also supported by the markedness constraint *[–high]/V[–stressed, μ]. As a result, these two constraints do not conflict.

As for IDENT(low) and IDENT(high), there are cases where the winning candidate is favoured by IDENT(low), and others where it is favoured by IDENT(high). In either case, however, the candidate is also supported by a higher-ranked constraint, and thus no conflict between IDENT(low) and IDENT(high) becomes decisive in the evaluation.

Now from (36), the descriptive generalisation (39) can be formulated:

(42) In unstressed syllables, short vowels that are in opposition with respect to [ATR] are prohibited. This requirement is enforced by the unfaithful mapping $[\pm ATR] \rightarrow [0 ATR]$, as in *morada* "adress" /mɔr-a-d-a/ → mu'ra.də vs. *mʊ'ra.də.

From this description, the following constraint hierarchies can also be posited:

***[±ATR]/V[–stressed, μ] >> IDENT(ATR)**

The definitions of the above constraints are as follows:

***[±ATR]/V[–stressed, μ]:** Assign one * for each unstressed short vowel that participates in an opposition with respect to [ATR].

IDENT(ATR): Assign one * for each change in the [ATR] feature between input and output.

The following tableaux (40) and (41) serve to confirm the ranking proposed above (note that the contextual specification /V[–stressed, μ] is omitted due to space constraints):

(43)	a	Id(lowVhigh)	*[+low]	*[–high]	Id(high)	*[+low]	Id(low)	*[±ATR]	Id(ATR)
	ɛ			*			*		
	a		*W	*		*W	L		
	ɔ			*			*	*	*
	i	*W		L	*W		*		

(44)	ɔ	Id(lowVhigh)	*[+low]	*[–high]	Id(high)	*[+low]	Id(low)	*[±ATR]	Id(ATR)
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ɪ				*				*
ɔ			*W	L			*W	L
o			*W	L			*W	*
ʊ				*			*W	L

Finally, from (37), the descriptive generalisation (42) can be formulated:

(45) In unstressed syllables, front short vowels specified as [–back] are prohibited. This requirement is enforced by an unfaithful mapping from [–back] to [+back] — for example, *dedada* /ded-ad-a/ → di'dadə vs. *di'dadə, and *pregar* /prɛg-a-r0/ → pri'ga.ɾ vs. *pri'ga.ɾ.

From this generalisation, the following ranking can be inferred:

***[–back]/V[–stressed, μ] >> IDENT(back)**

The definitions of the above newly introduced constraints are as follows:

***[–back]/V[–stressed, μ]**: Assign one * for each unstressed front short vowels.

IDENT(back): Assign one * for each change in the [back] feature between input and output.

The following tableau (43) serves to confirm the ranking proposed just above:

(46)

e	*[–back]/V[–stressed, μ]	IDENT(back)
ɪ		*
i	*W	L

The mapping of the winner vs. the loser is as follows:

/e/:[–low][–high][+ATR][–back] → [i]:[–low][+high][+ATR][+back]

/e/:[–low][–high][+ATR][–back] → * [i]:[–low][+high][+ATR][–back]

However, **the mapping [–back] → [+back] always occurs in tandem with the mapping [–high] → [+high], and never in isolation** — that is, /e ε/ → [i] vs. *[i], whereas /i/ → [i] vs. *[i]. Therefore, if we evaluate the output candidates [i] and *[i] for /i/ based on the previously established ranking ***[–back]/V[–stressed, μ] >> IDENT(back)**, *[i], which is favoured by the higher-ranked ***[–back]/V[–stressed, μ]**, would incorrectly win over [i], which is favoured by the lower-ranked IDENT(back), as shown in the tableau (44). This contradicts the observed facts.

(47)

i	*[–back]/V[–stressed, μ]	IDENT(back)
---	--------------------------	-------------

i	*	W
i	L	*

If we reverse, however, the ranking between $*[-\text{back}]/V[-\text{stressed}, \mu]$ and $\text{IDENT}(\text{back})$, adopting $\text{IDENT}(\text{back}) \gg *[-\text{back}]$, we can no longer account for the mapping $/e/ \rightarrow [i]$ vs. $*[i]$. Thus, **to derive the correct outputs**, it is necessary **to maintain the ranking $*[-\text{back}]/V[-\text{stressed}, \mu] \gg \text{IDENT}(\text{back})$** , while **positing an additional constraint ranked above $*[-\text{back}]/V[-\text{stressed}, \mu]$ that favours $/i/ \rightarrow [i]$ and disfavors $/i/ \rightarrow *[i]$** .

Among the output candidates for $/i/$, $[i]$ and $*[i]$ differ only in the value of the $[\text{back}]$ feature — the former being $[-\text{back}]$ and the latter $[\text{+back}]$ — while they share identical values for the remaining features: $[\text{+high}]$, $[-\text{low}]$, $[\text{+ATR}]$, and $[-\text{round}]$. Therefore, no markedness constraint pertaining to those shared features, even if ranked above $*[-\text{back}]$, can favour $[i]$ over $*[i]$: such constraints would support neither candidate. In fact, among all the markedness constraints extracted thus far, only $*[-\text{back}]/V[-\text{stressed}, \mu]$ expresses any preference between $[i]$ and $*[i]$; the remaining constraints — $*[\text{+low}]/V[-\text{stressed}, \mu]$, $*[-\text{high}]/V[-\text{stressed}, \mu]$, and $*[-\text{ATR}]/V[-\text{stressed}, \mu]$ — favour neither. Even the introduction of additional markedness constraints such as $*[-\text{round}]$ would yield the same result.

It is reasonable, thus, to assume that **the constraint ranked above $*[-\text{back}]$ must be some faithfulness constraint**. Until its specific content is determined, we shall refer to this constraint provisionally as F . When this ranking — $F \gg *[-\text{back}]/V[-\text{stressed}, \mu] \gg \text{IDENT}(\text{back})$ — is incorporated into the tableau discussed above, it yields the following structure (45) and (46) (note that no $*$ is assigned in the column for F , as its definition remains unspecified):

(48)

e	F	$*[-\text{back}]/V[-\text{stressed}, \mu]$	$\text{IDENT}(\text{back})$
i			*
i	(W)	*W	L

(49)

i	F	$*[-\text{back}]/V[-\text{stressed}, \mu]$	$\text{IDENT}(\text{back})$
i		*	W
i	W	L	*

The faithfulness constraint F must satisfy the following two conditions within the ranking $F \gg *[\text{back}]/V[-\text{stressed}, \mu] \gg \text{IDENT}(\text{back})$:

(50) F must **favour the mapping /i/ → [i] while disallowing /i/ → *[i]**.

(51) F must not disallow the mapping /e/ → [i]; that is, it must either **be neutral with respect to both /e/ → [i] and /e/ → *[i]**, or else actively **favour /e/ → [i]**.

Let us first consider the faithfulness violations involved in the mapping of /i/, with respect to condition (47). The mapping /i/ → [i] incurs no violations, whereas /i/ → *[i] involves a single faithfulness violation: the change from [–back] to [+back]. Therefore, the faithfulness constraint F can be taken as one that **prohibits the mapping [–back] → [+back], thereby disfavoring /i/ → *[i]**. However, since this constraint must outrank *[–back], it is evidently not identical to IDENT(back) — given that *[–back] >> IDENT(back).

Next, let us examine the faithfulness violations involved in the mappings of /e/, with respect to condition (48). The mapping /e/ → *[i] incurs a single violation: [–high] → [+high]. On the other hand, /e/ → [i] incurs two violations: [–high] → [+high] and [–back] → [+back]. That is, **/e/ → [i] entails more faithfulness violations than /e/ → *[i]**. It is therefore **unlikely that a faithfulness constraint F could favour /e/ → [i] while disallowing /e/ → *[i]**. Consequently, the most plausible assumption is that **the faithfulness constraint F supports neither /e/ → [i] nor /e/ → *[i]**.

Moreover, if the faithfulness constraint F were to target only the mapping [–back] → [+back] in /e/ → [i], and prohibit it, then F would effectively be IDENT(back), which would contradict our assumption that F must outrank *[–back] while IDENT(back) does not. Therefore, **the faithfulness constraint F must also be concerned with the shared mapping [–high] → [+high] found in both /e/ → [i] and /e/ → *[i], and likely serves to prohibit that change as well.**

To summarise the above discussion, **the faithfulness constraint F in question must simultaneously prohibit both the change [–back] → [+back] and [–high] → [+high]**. In order for such a constraint to penalise only the latter in the mapping /i/ → *[i], while assigning the same number of violations to both /e/ → *[i] and /e/ → [i], we can define F as **IDENT(high^back)**. The definition of this constraint is as follows:

IDENT(high^back): Assigns one violation mark * to a segment if there is a change in the value of either [high] or [back].

By ranking it as follows:

IDENT(high^back) >> *[–back]/V[–stressed, μ] >> IDENT(back)

we can coherently account for the mappings /i/ → [i] vs. *[i] and /e/ → [i] vs. *[i] without contradiction. Put simply, within the [−low] domain defined by the [±high] and [±back] features, any deviation — whether a single step (as in /i/ → [i] or /e ε/ → [i]) or a double step (as in /e ε/ → [i] → [i]) — results in an equal degree of violation. And when the same violations are incurred, the candidate that achieves a more optimal shape — namely, /e ε/ → [i] → [i] — emerges as the winner. The following tableaux (49) and (50) serve to confirm the ranking inferred above:

(52)

e	IDENT(high^back)	*[−back]/V[−stressed, μ]	IDENT(back)
i	*		*
i	*	*W	L

(53)

i	IDENT(high^back)	*[−back]/V[−stressed, μ]	IDENT(back)
i		*	
i	*W	L	*W

The introduction of the complex constraint IDENT(high^back) here serves not merely as a technical expedient, but as **a theoretical expression of a functional requirement of the system: the preservation of the underlying distinctive opposition between high front /i/ and mid front vowels /e ε/**. Thus, the ranking of IDENT(high^back) >> [−back]/V[−stressed, μ] should be interpreted as reflecting the system's resistance to neutralising this distinctive opposition. Without such a constraint, the system would favour maximal reduction to a reduced vowel [i] across the entire front vowel space /i e ε/, thereby collapsing a opposition that is both frequent and functionally load-bearing.

While the general vowel reduction pattern in unstressed syllables preserves the distinctive opposition between /i/ and /e ε/ through the ranking IDENT(high^back) >> [−back]/V[−stressed, μ], a more neutralised pattern is observed in final prosodically weakest syllables — namely, in final unstressed open syllables or in final unstressed syllables closed by [ʃ] or [ʒ], the only consonants specified as [−sonorant] in syllable coda — where even /i/ surfaces as [i], as shown in (51):

$$(54) \text{part}[\acute{i}:]mos \approx \text{part}[i]; \text{com}[\acute{e}:]mos \approx \text{com}[i]; \text{diss}[\acute{e}:]mos \approx \text{diss}[i]$$

(Note that *partimos* and *parte* are the first person plural and third person singular forms of the verb *partir* "to depart" in the present indicative; *comemos* and *come* are the first person plural

and third person singular forms of *comer* "to eat" in the present indicative; *dissemos* and *disse* are the first person plural and third person singular forms of *dizer* "to say" in the imperfect subjunctive.) This neutralisation may be accounted for by a positional markedness constraint $*[-\text{back}]/V[-\text{stressed}, \mu] __ ([-\text{sonorant}])\#$, which dominates $\text{IDENT}(\text{high} \wedge \text{back})$ specifically in these environments. The definition of this constraint is as follows:

$*[-\text{back}]/V[-\text{stressed}, \mu] __ ([-\text{sonorant}])\#$: Assign one * for each unstressed short front vowel that occurs in a final open syllable or in a final syllable closed by a $[-\text{sonorant}]$ segment.

The meaning of this constraint is that, in the weakest unstressed position at word-final edge (cf. Mattoso Câmara 1972), the underlying opposition between /i/ and /e ε/ can no longer be maintained. In Brazilian Portuguese as well, in this same weak unstressed position, the opposition between /i/ and /e ε/ is likewise neutralised, yielding /i e ε a o o u/ → [i i i ɐ u u u]; by contrast, in other unstressed positions, the opposition is preserved, as seen in /i e ε a o o u/ → [i e e a o o u].

Now let us turn to the problem of the long vowels. In this stratum, the long vowels [ε: a: o:] undergo no change in either quality or quantity. This suggests that the presumed inviolable constraint **IDENT-QUANTITY** is at work, and that the faithfulness constraints $\text{IDENT}(\text{high})$, $\text{IDENT}(\text{low})$, $\text{IDENT}(\text{ATR})$, $\text{IDENT}(\text{back})$, and $\text{IDENT}(\text{round})$ dominate all markedness constraints related to [high], [low], [ATR], [back], and [round] features — with the exception of $*[+\text{low}]/V[-\text{stressed}, \mu]$, $*[-\text{high}]/V[-\text{stressed}, \mu]$, $*[-\text{ATR}]/V[-\text{stressed}, \mu]$ >> $\text{IDENT}(\text{ATR})$, and $*[-\text{back}]/V[-\text{stressed}, \mu]$. The definition of the above presumed inviolable constraint is as follows:

IDENT-QUANTITY: Assign one * per mora difference between input and output — e.g. a difference of 1 mora incurs 1 *, a difference of 2 mora incurs 2 *, etc.

To summarise, in order to derive the mapping i e ε a o o u ε: a: o: → i i i ɐ u u u ε: a: o:, it suffices to have the following constraint interactions:

(55) The inviolable constraint **IDENT-QUANTITY**;

(56) $\text{IDENT}(\text{low} \vee \text{high})$ >> $*[+\text{low}]/V[-\text{stressed}, \mu]$ >> $*[-\text{high}]/V[-\text{stressed}, \mu]$ >> $\text{IDENT}(\text{high})$

(57) $*[+\text{low}]/V[-\text{stressed}, \mu]$ >> $\text{IDENT}(\text{low})$

(58) $*[\pm ATR]/V[-stressed, \mu] \gg IDENT(ATR)$

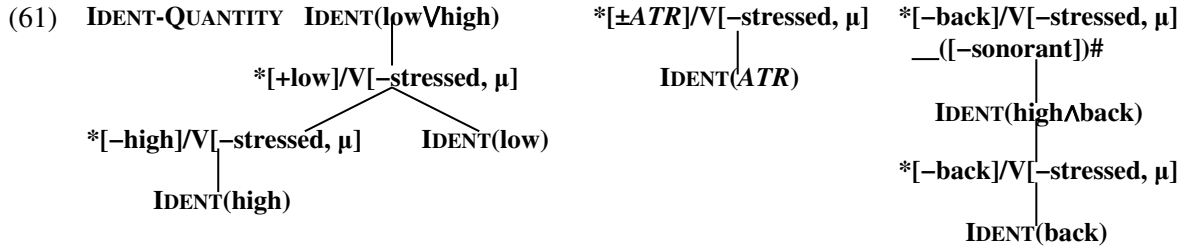
(59) $*[-back]/V[-stressed, \mu] _ ([-sonorant])\# \gg IDENT(high\wedge back) \gg *[-back]/V[-stressed, \mu] \gg IDENT(back)$

(60) $IDENT(high), IDENT(low), IDENT(ATR), IDENT(round) \gg * [+high], * [-high], \text{etc.}$

It is evident from the following set of mappings that $[+high]$, $[-ATR]$, and $[-back]$ do not conflict with each other, and likewise that $[+low]$ and $[-back]$ do not conflict either (features whose values change are shaded):

i: $[-low][+high][+ATR][-back][-round] \rightarrow i: [-low][+high][0 ATR][-back][-round]$
 e: $[-low][-high][+ATR][-back][-round] \rightarrow i: [-low][+high][0 ATR][+back][-round]$
 ε: $[-low][-high][-ATR][-back][-round] \rightarrow i: [-low][+high][0 ATR][+back][-round]$
 a: $[+low][-high][+ATR][+back][-round] \rightarrow \text{e: } [-low][-high][0 ATR][+back][-round]$
 ɔ: $[-low][-high][-ATR][+back][+round] \rightarrow u: [-low][+high][0 ATR][+back][+round]$
 o: $[-low][-high][+ATR][+back][+round] \rightarrow u: [-low][+high][0 ATR][+back][+round]$
 u: $[-low][+high][+ATR][+back][+round] \rightarrow u: [-low][+high][0 ATR][+back][+round]$

The Hasse diagrams for the constraint hierarchies in (52) – (57) are as shown in (58):



4.2. Postlexical Stratum 2: Vocalic Quantity Adjustment

A descriptive generalisation regarding vowel quantity alternations in Contemporary EP may be formulated as in (59):

(62) Long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. a:'tivo $\rightarrow$ a'ti:vu, *a:'ti:vu), suggesting the ranking $*\mu\mu/V \gg MAX(\mu)$,

except when the vowel occurs in a stressed syllable, where short vowels are prohibited.

This requirement is enforced by epenthesising one mora if the input is a short vowel (e.g. ata /ata/ $\rightarrow$ ['a:tə] vs. *['atə]; pasta /pasta/ $\rightarrow$ ['pa:ʃtə] vs. *['paʃtə]), suggesting the

ranking $*\sigma[\mu] \gg \text{DEP}(\mu)$, or by maintaining the length if the input is a long vowel (e.g. *acta* /a:ta/ → ['a:tə] vs. *['atə]; /mɛ:stre/ → ['mɛ:ʃtri] vs. *['mɛʃtri]), suggesting the ranking $*\sigma[\mu] \gg *_{\mu\mu}/V$,

except when the vowel occurs in a closed syllable ending in [t], [r], [ɪ] or [ʊ], where long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. /nɔ:rma/ → ['nɔrmə], *['nɔ:rmə]; /nɔ:rmal/ → [nɔr'maɪ], *[nɔ:r'maɪ]), suggesting the ranking $*_{\mu\mu\mu} \gg *_{\sigma[\mu]}$.

The constraint rankings (60) and (61) may be derived from the above descriptive generalisation:

(63) $*_{\mu\mu\mu} \gg *_{\sigma[\mu]} \gg *_{\mu\mu}/V \gg \text{MAX}(\mu)$

(64) $*_{\mu\mu\mu} \gg *_{\sigma[\mu]} \gg *_{\mu\mu}/V, \text{DEP}(\mu)$

The definitions of the above constraints are as follows:

$*_{\mu\mu\mu}$: Assign one * for each syllable with three moras.

$*_{\sigma[\mu]}$: Assign one * for each–stressed syllable with only one mora.

$*_{\mu\mu}/V$: Assign one * for each vowel linked to two moras.

$\text{MAX}(\mu)$: Assign one * for each mora deleted.

$\text{DEP}(\mu)$: Assign one * for each mora epenthesised.

On the other hand, syllable quantity adjustment can also be achieved not through moraic operations but by the epenthesis or deletion of segments. For example, in mappings such as *ata* " " /a:ta/ → ['a:tə] vs. *['aṭtə], or *norma* "" /nɔ:rma/ → ['nɔrmə] vs. *['nɔ:mə], the asterisked candidates *['aṭtə] and *['nɔ:mə] are eliminated due to violations of DEP and MAX, respectively. The definitions of these constraints are as follows ("segment" here refers to a root node in the quality feature hierarchy abstracting away from quantity; therefore, deletion or insertion of moras alone is not considered a violation of DEP or MAX):

DEP: Assign one * for each segment epenthesised.

MAX: Assign one * for each segment deleted.

First, in the mapping *ata* "minutes, record" /a:t-a/ → ['a:tə] vs. *['aṭtə], the winner ['a:tə] is favoured by DEP, while the loser *['aṭtə] is favoured by $*_{\mu\mu}/V$, which indicates the ranking $\text{DEP} \gg *_{\mu\mu}/V$. Note that both candidates violate none of the constraints $*_{\mu\mu\mu}$, $*_{\sigma[\mu]}$, MAX,

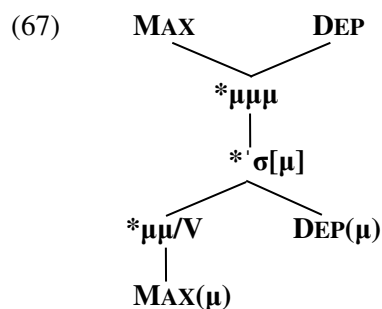
MAX(μ), or DEP(μ), and thus are equally evaluated with respect to these constraints. Next, in the mapping /a:t-a/ $\rightarrow$ ['a:.tə] vs. *[tə], the winning candidate ['a:.tə] is favoured by MAX, whereas the losing candidate *[tə] is favoured by $*\mu\mu/V$, which supports the ranking **MAX $\gg$ $*\mu\mu/V$** . Note also that ['a:.tə] satisfies MAX(μ), while *[tə] violates it; however, since $*\mu\mu/V \gg$ MAX(μ), candidate evaluation by Max(μ) does not affect the evaluation determined by $*\mu\mu/V$. Since MAX and DEP do not conflict in syllable quantity adjustment, the following constraint ranking (62) can be derived from the above discussion:

(65) MAX, DEP $\gg$ $*\mu\mu/V$

Next is the ranking among MAX, DEP and $*\mu\mu\mu$, all of which are ranked above $*\mu\mu/V$. With the exception of word-final trimoraic syllables — such as [-'me:r] in *comer* /kɔm-e-r/ $\rightarrow$ [ku'me:r] "to eat" — which require special theoretical treatment and which shall be discussed below, trimoraic syllables are found only in loanwords, as in *e-mail* /imeil/ $\rightarrow$ [i'meɪl̩] "e-mail", with the final syllable [-'meɪl̩]. In such cases, neither epenthesis — as in *[i.meɪ.li] — nor deletion — as in *[i'meɪ] — is attested, which suggests, at least provisionally, the ranking (63):

(66) MAX, DEP $\gg$ $*\mu\mu\mu$.

The Hasse diagrams for the constraint hierarchies in (60) - (63) are as shown in (64):



Let us return to the trimoraic [-'me:r] in *comer* [ku'me:r]. Considering the process of stress assignment and assuming the underlying representation of *comer* to be /kɔm-e-r0/ — where /0/ denotes a floating mora lacking segmental content (cf. Section 6.6) — it is more explanatory to posit that, in Postlexical Stratum 3, the input [ku'me:ɾ] to this stratum yields the output [ku'me:ɾ] within the same stratum, as follows:

comer /kɔm-e-r0/ $\rightarrow$ [kɔ.me.ɾ] $\rightarrow$ [kɔ'me.ɾ] $\rightarrow$ [ku'me.ɾ] $\rightarrow$ [ku'me:ɾ]

cf. *comera* /kɔm-e-ra/ → [kɔ.me.ra] → [kɔ'me.ra] → [ku'me.rɐ] → [ku'me:.rɐ]

Subsequently, resyllabification in a later stratum produces the final surface form [ku'me:r]. In this stratum, the ranking *C, MAX(μ), MAX, DEP >> *μμμ accounts for the exclusion of ill-formed outputs such as *ku'mer, *ku'me:.ri, and *ku'me:. Under this analysis, the output of /kɔm-e-r0/ at this stratum is ku'me:.ɾ, which does not violate the constraint *μμμ.

The following tableaux (65) – (72) serve to test the constraint ranking established above:

(68) *ativo* /a:t-iv-o/

a:'tɪvʊ	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	'σ[μ]	DEP(μ)
☞ a:'tɪ:vʊ					*	*		*
a:'tɪ:vʊ					**W	L		*
a:'tɪvʊ				*W	*	L	*W	L
a:'tɪrvʊ		*W			L	*		*
'tɪ:vʊ	*W				*	**W		*

(69) *ata* /at-a/

'atɐ	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	'σ[μ]	DEP(μ)
☞ 'a:tɐ					*			*
'atɐ				*W	L		*W	L
'artɐ		*W			L			*
'a:tɐ			*W		*			**W

(70) *ata* /a:ta/

'a:tɐ	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	'σ[μ]	DEP(μ)
☞ 'a:tɐ					*			
'atɐ				*W	L	*W	*W	
'artɐ		*W			L			
'a:tɐ			*W		*			*W

(71) *mestre* /mɛ:stre/

'mɛ:ʃtɾi	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	'σ[μ]	DEP(μ)
☞ 'mɛ:ʃtɾi					*			
'mɛʃtɾi				*W	L	*W	*W	

(72) *norma* /nɔ:rma/

'nɔ:r.mɐ	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	'σ[μ]	DEP(μ)
☞ 'nɔ:r.mɐ						*		
'nɔ:r.mɐ			*W		*W	L		
'nɔ:mɐ	*W				*W	*		

(73) *normal* /nɔ:rmaɪ0/

nɔ:r'ma:l	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
ɹ̥nɔ:r'ma:l					*	*		*
nɔ:r'ma:l			*W		**W	L		*
nɔ:r'ma:l			*W	*W	*	L	*W	L
nɔ:'ma:l	*W				**W	*		*
nɔ'ma:l	*W				*	**W		*

(74) *past* /pa:stə/

'paste	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
ɹ̥'pa:stə					*			*
'pa:stə				*W	L		*W	L
'pa:tə	*W				*			
'partə		*W			L			*W

(75) *comer* /kɔmer0/

ku'me:f	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
ɹ̥ku'me:f								
ku'me:f								
ku'me:								

In Contemporary EP, the constraint ranking outlined above ensures that underlying long vowels retain their length faithfully only in stressed open syllables and in stressed closed syllables whose coda is [ʃ] or [ʒ]. In these contexts, furthermore, underlying short vowels are also realised as long vowels due to the ranking $*'σ[μ] \gg *μμ/V \gg MAX(μ)$, which results in a surface-level neutralisation of vowel length opposition. The only context where this opposition could potentially remain is in unstressed syllables; even there, however, $*μμ/V \gg MAX(μ)$ enforces surface-level neutralisation of length. In such cases, the underlying opposition is preserved through vowel quality reduction: only underlying short vowels undergo weakening in vowel quality. Thus, since **the underlying opposition in vowel length is either neutralised on the surface or reinterpreted as a quality opposition**, it is therefore likely that **native speakers of EP do not perceive the opposition in vowel length as of distinctive nature**.

5. Analysis Summary

6. Theoretical Applications

In this section, we shall examine several phenomena that can be accounted for by appealing to (i) the opposition between underlying short vowels $\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\text{ɔ}} \overset{\mu}{o} \overset{\mu}{u}$ and underlying long vowels $\overset{\mu\mu}{e} \overset{\mu\mu}{a} \overset{\mu\mu}{\text{ɔ}}$, as posited in Section 3.2., and (ii) the postlexical phonological processes discussed in Section 4, namely vowel quality reduction and vowel quantity adjustment, both of which are sensitive to stress and take place across the two strata of the postlexical level.

6.1. Lack of Quality Reduction of Nasal Vowels in Unstressed Syllables

Contemporary EP typically exhibits five distinct nasal monophthongs [ĩ ẽ ã õ û] at the phonetic level. These vowels undergo quantity alternation — [ĩ: ẽ: ã: õ: û:] $\approx$ [ĩ ẽ ã õ û] — between stressed and unstressed syllables, but do not undergo the kind of vowel quality reduction characteristic of underlying short oral vowels, as seen in $p[\acute{e}:]sa \approx p[i]samos$ from the verb *pesar* "to weigh" and $m[\acute{o}:]ra \approx m[u]ramos$ from the verb *morar* "to live". For example, the verb *pensar* "to think" shows the alternation $p[\acute{e}:]sa \approx p[\tilde{e}]samos$, but not $p[\acute{e}:]sa \approx *p[\tilde{i}]samos$; similarly, the verb *contar* "to tell; to count" exhibits the alternation $c[\acute{o}:]ta \approx c[\tilde{o}]tamos$, but not $c[\acute{o}:]ta \approx *c[\tilde{u}]tamos$. Why, then, do these vowels not undergo quality reduction? **The fact that, in alternations between stressed and unstressed vowels, only vowel quantity changes while quality remains constant is a property shared with the underlying long vowels** — $/\varepsilon: a: \text{ɔ}:/ \rightarrow [\acute{e}: \acute{a}: \acute{o}:] \approx [\varepsilon a \text{ɔ}]$. But does this mean that nasal vowels are underlyingly long vowels?

When addressing phonological sequences involving nasal vowels, the initial problem lies in the phonological interpretation of the nasal vowel itself. In the case of Portuguese, there exist two competing analyses: one views the nasal vowel as "a single segment", while the other considers it as "a phonetic realisation of a sequence of two segments". The former is represented by works such as Lüdke (1953), but is no longer a widely accepted view today. The latter, by contrast, branches into two theoretical traditions: (i) a functionalist–structuralist interpretation, according to which the nasal vowel is the phonetic realisation of a sequence composed of a vocalic phoneme and an archiphoneme /N/ (Barbosa 1994, etc.), and (ii) an autosegmental account, in which nasal vowels derive phonologically from a sequence consisting of a vocalic segment and an autosegment [+nasal] (Mateus and Andrade 2002; Mateus et al. 2003, etc.). This study aligns with the latter two "two-segment sequence" analyses, and argues — on the basis of the three considerations presented below — that the phonetically realised simple nasal vowel [Ń] in Portuguese is phonologically /Vn/; in other words, a sequence consisting of a vocalic segment

/V/ followed by a nasal consonantal segment /n/.

The first piece of evidence lies in the phonological alternation $[Vn] \approx [\tilde{V}]$ observed within the same morpheme under different phonological environments. For instance, the alternation $[i.na-] \approx [\tilde{i}.s\epsilon-]$ of the negative prefix *in-* in *in+ativo* $[i.na'ti:vu]$ "inactive" vs. *in+sanável* $[\tilde{i}.s\epsilon'na:v\epsilon\lambda]$ "incurable" (where + marks the boundary between a derivational prefix and the base), and the alternation $[\underline{i}.m\epsilon'na:-] \approx [\underline{i}.m\tilde{\epsilon}'da:-]$ of the root morpheme *irman-* in *irman+ar* $[\underline{i}.m\epsilon'na:r]$ "to join together" vs. *irman+dade* $[\underline{i}.m\tilde{\epsilon}'da:\delta i]$ "brotherhood", suggest the following: an underlying sequence /Vn/ is syllabified as $\{V\}\{nV\}$ before another vowel /V/, whereas before a consonant–vowel sequence /CV/ it is syllabified as $\{Vn\}\{CV\}$, where $\{ \}$ indicates syllabic parsing. In the former case $\{V\}\{nV\}$, the underlying /n/ is syllabified as the onset of the following syllable and is phonetically realised as [n]. In the latter $\{Vn\}\{CV\}$, /n/ is syllabified as the coda of the preceding syllable, nasalises the preceding vowel /V/, and is then deleted at the phonetic level — $\{Vn\}\{CV\} \rightarrow \{\tilde{V}n\}\{CV\} \rightarrow \{\tilde{V}_\}\{CV\} \rightarrow \{\tilde{V}:\}\{CV\}$. As illustrated by American English *cannot* $['kænɑ:t] \approx can't$ $['kænt]$, it is a cross-linguistically common phenomenon for a vowel to become nasalised before a nasal coda. Moreover, as seen in certain dialects of American English where *can't* is realised as $[kæ:t]$, the nasal coda [n] may be deleted after nasalising the preceding vowel (cf. Kenstowicz 1994: 14). These observations suggest that the interpretation proposed above is not phonologically implausible.

The second piece of evidence concerns the phonetic realisation of the consonant /r/. This segment is realised as a tap [ɾ] when it follows an oral vowel, i.e. $/VrV/ \rightarrow [VrV]$ — as in *caro* "expensive" $/kar-o/ \rightarrow ['ka:ru]$ — but as a trill [ʀ] or a fricative [ʁ] when it follows another consonant, such as /s/, /l/, or /t/, i.e. $/Vs_rV/, /Vl_rV/, /Vt_rV/ \rightarrow [VʒʀV], [VlʀV], [VʀV]$ — as in *Israel* $/isra\epsilon l/ \rightarrow [iʒʀ\epsilon' \epsilon\lambda]$, *melro* "blackbird" $/m\epsilon lro/ \rightarrow ['m\epsilon lʀu]$, and *carro* "car" $/karro/ \rightarrow ['ka:ʀu]$. This distribution shows that when /r/ occurs in syllable-initial position, it is realised as [ɾ] if preceded by a [–consonantal] segment, and as [ʀ] if preceded by a [+consonantal] one. Now, since /r/ is also realised as [ʀ] following a nasal vowel — e.g. $[\tilde{V}rV]$ as in *genro* $['ʒ\tilde{\epsilon}:ru]$ — it is reasonable to assume that, phonologically, a [+consonantal] segment intervenes between the vowel and /r/. That is, **the sequence $[\tilde{V}rV]$ should be analysed phonologically as $/VnrV/$.**

The third piece of evidence concerns the phonetic realisation of the consonants /b d g/. In Contemporary the European variety of Portuguese, it is generally the case that in the phonic environment $[V_V]$, these segments assimilate the [+continuant] feature of the preceding vowel and are realised as fricatives $[\beta \delta \gamma]$ — e.g. $/VbV/, /VdV/, /VgV/ \rightarrow [V\beta V], [V\delta V], [V\gamma V]$, as in *o boi* $[u'\beta oi]$ "the ox", *o dia* $[u'\delta i:v]$ "the day", *o gato* $[u'\gamma a:tu]$ "the cat". By contrast, in the

environment $[\tilde{V}_V]$, these same consonants are invariably realised as stops $[b\ d\ g]$ — e.g. $[\tilde{V}]/bV/$, $[\tilde{V}]/dV/$, $[\tilde{V}]/gV/ \rightarrow [\tilde{V}bV]$, $[\tilde{V}dV]$, $[\tilde{V}gV]$, as in *um boi* $[\tilde{u}'boi]$ "an ox", *um dia* $[\tilde{u}'di:\varepsilon]$ "a day", *um gato* $[\tilde{u}'ga:tu]$ "a cat". This pattern supports the analysis that $[\tilde{V}_V]$ is phonologically $/Vn_V/$, and that $/n/$, unlike vowels, is $[-\text{continuant}]$.

Consequently, $/b\ d\ g/$ do not assimilate a $[+\text{continuant}]$ feature from the preceding $/n/$ — which does not bear this value — and their underlying $[-\text{continuant}]$ specification is preserved in surface realisation (for example, *longe* $/lonʒe/$ "far (away)" is phonetically realised as $['lõ:ʒi]$, and not as $*['lõ:\text{dʒi}]$; this indicates that the consonant following $/n/$ — in this case, $/ʒ/$, which is $[+\text{continuant}]$ — does not assimilate the $[-\text{continuant}]$ feature of $/n/$; it follows, then, that $/b\ d\ g/$ are underlyingly $[-\text{continuant}]$ stops, since they cannot assimilate the $[-\text{continuant}]$ feature of $/n/$). From these three points, the interpretation that "the phonetically nasalised vowel $[\tilde{V}]$ is derived from the underlying segmental sequence $/Vn/$ " may be regarded as phonologically well grounded.

From the above discussion, two main points (73) and (74) have emerged:

(76) Nasal vowels exhibit the same type of alternation as underlying long vowels. That is, in alternations between stressed and unstressed syllables, vowel quantity changes but vowel quality remains constant — namely, $[\acute{í}: \acute{é}: \acute{ê}: \acute{ó}: \acute{ú}:] \approx [\tilde{í} \tilde{e} \tilde{e} \tilde{o} \tilde{u}]$, just as the underlying long vowels $/\varepsilon: a: \text{ɔ}:/ \rightarrow [\acute{é}: \acute{á}: \acute{ó}:] \approx [\varepsilon\ a\ \text{ɔ}]$.

(77) Nasal vowels are the phonetic realisation of a segmental sequence consisting of a vocalic segment followed by a nasal consonant segment, i.e., $/Vn/$.

What follows from this is the following: **nasal vowels, at the input to the vowel quality modification and quantity adjustment tiers, are long vowels of bimoraic weight** — namely, $\acute{í}: \acute{é}: \acute{ê}: \acute{ó}: \acute{ú}:$ — just like the underlying long vowels — $\varepsilon: a: \text{ɔ}:$. As derived long vowels, they are subject to the same constraint hierarchy and receive the same treatment as underlying long vowels with respect to both quality and quantity. Precisely because they are bimoraic, they **undergo no change in quality at the vowel quality modification tier** — just like underlying long vowels — and then, **at the quantity adjustment tier**, they are **realised with full quantity in stressed syllables**, but **lose one mora in unstressed syllables**, undergoing shortening in the output — $\acute{í}: \acute{é}: \acute{ê}: \acute{ó}: \acute{ú}: \approx \tilde{í}: \tilde{e}: \tilde{e}: \tilde{o}: \tilde{u}: \rightarrow \acute{í}: \acute{é}: \acute{ê}: \acute{ó}: \acute{ú}: \approx \tilde{í} \tilde{e} \tilde{e} \tilde{o} \tilde{u}$, just as $\acute{é}: \acute{á}: \acute{ó}: \approx \varepsilon: a: \text{ɔ}: \rightarrow \acute{é}: \acute{á}: \acute{ó}: \approx \varepsilon\ a\ \text{ɔ}$. On the other hand, since they are underlyingly segmental sequences $/Vn/$, **nasalisation of the Vn sequence must occur** prior to the vowel quality modification tier — that is, **between the syllabification tier and the quality modification tier**. Furthermore,

insofar as they surface as long vowels, the /Vn/ sequence must be bimoraic, and **no mora deletion takes place during the process of nasalisation.**

$\begin{smallmatrix} \mu & \mu \\ /Vn/ \end{smallmatrix}$		$\begin{smallmatrix} \mu & \mu \\ /V/ \end{smallmatrix}$		Input representation
in en ɛn an ɔn on un	in en ɛn an ɔn on un	ɛ: a: ɔ:	ɛ: a: ɔ:	Syllabification Tier
ín éñ én án óñ ón ún	—	é: á: ó:	—	Stress Assignment Tier
ĩ: ê: ê: ẽ: õ: õ: ũ:	ĩ: ê: ê: ẽ: õ: õ: ũ:	—	—	Nasalisation Tier
—	—	—	—	Vowel Quality Modification Tier
—	ĩ ẽ ẽ ẽ õ õ ũ	—	ɛ a ɔ	Vowel Quantity Adjustment Tier
ĩ: ê: ê: ẽ: õ: õ: ũ:	ĩ ẽ ẽ ẽ õ õ ũ	é: á: ó:	ɛ a ɔ	Output representation

At the nasalisation tier, a mapping such as in, en, ɛn, an, ɔn, on, un → ĩ:, ê:, ê:, ẽ:, õ:, õ:, ũ takes place. In particular, the mappings en, an, ɔn → ê:, ẽ:, õ: resemble the weakening of unstressed short vowels and may likewise be analysed as a kind of reduction. Because expiratory airflow is partially consumed by nasal resonance, the volume of airflow reaching the oral cavity is reduced, which results in the neutralisation of vowel quality oppositions.

Expected Constraints in this mapping are likely to be as follows:

IDENT-QUANTITY

*[−ATR]/__N]σ >> IDENT(ATR)

*[+low]/__N]σ >> IDENT(low)

IDENT[low∨ATR] (as for this constraint, it must be examined whether simultaneous changes in [low] and [ATR] are prohibited or not.)

*VN >> MAX(C)

The constraint hierarchy is as follows:

IDENT[low∨ATR] >> *[+low]/__N]σ >> *[-ATR]/__N]σ >> IDENT(ATR)

*[+low]/__N]σ >> IDENT(low)

*VN >> MAX(C)

6.2. Diphthongisation of vowel segments before palatal consonants

In Contemporary EP, as Barbosa (1988: 346) points out, "hoje em dia, se encontra com frequência um *iod* entre qualquer vogal acentuada e qualquer uma das quatro consoantes palatais indicadas inicial de da sílaba seguinte, e isso quer tal *iod* possa quer não possa explicar-se etimologicamente, quer tenha quer não tenha representação gráfica: palavras como *caixa*, *graxa*, *olhos*, *hoje*, *unha*, *beijo*, *velho* são pronunciadas com ou sem [i], mais ou menos nítido, e isto varia de idiolecto para idiolecto e, por vezes, no mesmo idiolecto". The terms *as quatro consoantes palatais* and *iod* refer, in IPA terms, to [ʃ ʒ ʎ ɲ] and [i], respectively. In orthographic terms, these are represented as *x*, *ch* for [ʃ]; *j*, *g* (before *i*, *e*) for [ʒ]; *lh* for [ʎ]; and *nh* for [ɲ]. The glide [i], when graphically represented, appears as *i*.

For instance, in the third-person singular forms of the verbs *relaxar* "to relax" and *taxar* "to tax" in the present indicative, the stressed vowel, spelled *a* in both cases, exhibits variation between a monophthong and a diphthong ending in [i]: *rel[á:]xa* $\approx$ *rel[ái]xa* and *t[á:]xa* $\approx$ *t[ái]xa*. What is particularly interesting is the realisation of this radical vowel *a* in unstressed syllables. In the former case, the same type of variation between a monophthong and a diphthong — *rel[a]xar* $\approx$ *rel[ai]xar* — is again observed in the unstressed syllable, just as in the stressed one. In contrast, in the latter case, only the monophthong is attested: *t[ɐ]xar*. What does this imply? (The following is a reinterpretation of Makino (2006:) within the theoretical framework adopted in this study.)

The alternation between the stressed and unstressed vowels without [i] in these verbs — namely, *rel[á:]xa* $\approx$ *rel[a]xar* and *t[á:]xa* $\approx$ *t[ɐ]xar* — reflects the alternation of a underlying long vowel /^μa/ in the former, and that of an underlying short vowel /^μa/ in the latter. Based on this, we assume the underlying representations of these verbs to be *relaxar* /rel^μaʃ-a-r0/ and *taxar* /t^μaʃ-a-r0/, respectively.

Next, let us turn to the behaviour of [i]. The variation or alternation between a monophthong — here abbreviated as V — and a diphthong — abbreviated as V_i — is limited to open syllables preceding the consonants [ʃ ʒ ʎ ɲ], and does not occur before [s z l n], which are adjacent to them in the place of articulation feature within the consonant system. It is therefore clear that the [i] element in V_i is in some way conditioned by articulatory properties that are present in [ʃ ʒ ʎ ɲ] but absent in [s z l n]. From a diachronic perspective, many of these consonants appear to have arisen through coalescence involving mutual assimilation between alveolar consonants such as [s z l ɲ] and the glide [j] or [i]. Examples include lat. *rŭssĕu-* > pt. *ro[ʃ]o*; lat. *cervĕsĭa-* > pt. *cerve[ʒ]a*; lat. *sĕdĕa* > pt. *se[ʒ]a*; lat. *allĭu-* > pt. *a[ʎ]o*, lat. *tĕnĕō* > te[n]o (cf. Williams 1961: 42, 44, 46, 50, 91). A similar process is observed in contemporary English, as in *I miss you* [ʌ]

mis ju], which can phonetically surface as [Λ₁ mɪʃu] due to the coalescence of [-s j-] into [-ʃ-] via mutual assimilation.

In Contemporary EP, the underlying segments /s z ʎ n/ are generally interpreted as simple segments with a single place of articulation. However, in the present study — taking into account both their historical development and their articulatory characteristics in the contemporary language — these palatal consonants are analysed as complex segments involving two distinct oral constrictions: a primary articulation formed by the tongue crown against the alveolar ridge — a gesture shared with /s z l n/, represented as **Coronal-[+anterior]** node in feature geometry (cf. Kenstowicz 1994: 136–168); and a secondary articulation formed by the tongue front against the hard palate — a gesture characteristic of [j ɨ], represented as **Dorsal-[-back, +high]** node. If phonological features are understood as neural commands issued by the brain to the speech organs in order to execute specific muscular gestures (cf. Kenstowicz 1994: 20, 138), then in the case of /ʃ z ʎ n/, one may say that two distinct constriction commands co-occur at the phonological level: a command to the tongue crown to constrict against the alveolar ridge, and a command to the tongue body to constrict against the hard palate. These co-occurring commands give rise, at the phonetic level, to articulations that range from post-alveolar to pre-palatal to palatal (for the place of articulation of [ʃ z ʎ n], cf. Barbosa 1994¹: 63).

The above discussion can be summarised in the following two points:

(78) The underlying representations of the verbs *relaxar* and *taxar* are /relaʃ^μ-a-r0/ and /taʃ^μ-a-r0/, respectively.

(79) The consonants /ʃ z ʎ n/ are analysed as complex segments involving two simultaneous oral gestures: a coronal constriction at the alveolar ridge — corresponding to the Coronal-[+anterior] node — and a dorsal constriction at the hard palate — corresponding to the Dorsal-[-back, +high] node.

With the assumptions in (75) and (76), it becomes possible to account for the variation observed in *rel*[á:]*xa* ≈ *rel*[áĩ]*xa* ≈ *rel*[a]*xar* ≈ *rel*[aĩ]*xar* and *t*[á:]*xa* ≈ *t*[áĩ]*xa* ≈ *t*[e]*xar*. Let us begin with the mapping /relaʃ^μ-a/ ≈ /relaʃ^μ-a-r0/ → *rel*[á:]*xa* ≈ *rel*[a]*xar*. Here, the stressed vowel [á:] is selected as the optimal output corresponding to the input underlying long vowel /a^μ/, evaluated as the least marked candidate under the constraint ranking in (64). The unstressed vowel [a] is similarly derived. In contrast, in the mapping /relaʃ^μ-a/ ≈ /relaʃ^μ-a-r0/ → *rel*[áĩ]*xa* ≈ *rel*[aĩ]*xar*, the vowel quality /a/ of the underlying long vowel /a^μ/ is first delinked from the second mora, resulting in /a^μ/. Then, to satisfy the vocalic constriction required by a syllable nucleus, a floating

[−consonantal, +sonorant]–Oral node is introduced and linked to the delinked mora. Finally, the Dorsal[−back, +high] node — the secondary palatal articulation inherent in /ʃ/ — spreads onto this node. The resulting complex — /aʃ/ → a^μ ʃ^μ → aʃ^μ — is then evaluated under the constraint ranking in (64), and its optimal phonetic realisations emerge as [áɿ] and [aɿ] as in (77).

(80) *relaxa* /relaʃ-a/

ri'la:ʃe	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
ri'la:ʃe					*			
ri'laɿʃe					*			
ri'larʃe		*W			L			
ri'laʃe				*W		*W		

Next, consider the mapping /taʃ-a/ → t[á:]xa. In this case, the vowel quality /a/, which is associated with only one mora in the underlying representation /a^μ/, becomes associated with an additional mora under stress in Postlexical Stratum 3, resulting in a long vowel [á:]. This long vowel is then selected as the optimal output corresponding to the input underlying short vowel /a^μ/, as it is evaluated to be the least marked candidate under the constraint ranking given in (64). On the other hand, in the mapping /taʃ-a/ → t[áɿ]xa, the vowel quality /a/ of the underlying short vowel /a^μ/ is not linked to the additional mora assigned under stress and is realised as a short vowel [á]. In its place, an [−consonant, +sonorant]–Oral node — introduced to satisfy the syllable nucleus's requirement for vocalic constriction — is linked to that mora. Finally, the Dorsal[−back, +high] node, corresponding to the secondary palatal articulation of /ʃ/, spreads onto this Oral node. The resulting segment sequence is then selected as the optimal output — [áɿ] — under the constraint ranking given in (64). The candidate evaluation tableau in this case is given in (78):

(81) *taxa* /taʃ-a/

'taʃe	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
'ta:ʃe					*			*
'taɿʃe					*			*
'tarʃe		*W			L			*
'taʃe				*W	L			L

Finally, in the mapping /taʃ-a/ → t[ɐ]xar, the segment [ɐ], which is the output of the underlying

short vowel /a/ in Postlexical Stratum 2 and serves as the input to Postlexical Stratum 3, occupies an unstressed syllable position. As such, it neither receives an additional mora nor associates with one. It is, therefore, selected as the optimal output under the constraint ranking (64), as shown in (79):

(82) *taxar* /ta^μf-a-r0/

te'fa.ɾ	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)	*'σ[μ]	DEP(μ)
te'fa.ɾ					*			*
te'fa.ɾ					*			**W
te'fa.ɾ				*W	L		*W	L

It is worth noting that, regarding this palatal extension, one of the informants cited in Makino (2005) made the particularly intriguing observation that "after speaking for a long time and becoming tired, I tends to say *rel*[a_i]*xar* instead of *rel*[a]*xar*". Phonetically, this phenomenon may be explained by a form of regressive assimilation that facilitates the articulatory transition from /a/ or /a/ to /f/.

6.3. Phonetic Realisation of Vowel Coalescence in Phrasal Phonology

It was already noted in Section 3.2 that the [a] in the imperative sentence *Fic*[a]*qui* (*Fica aqui*) "Stay here", where the locative adverb *aqui* follows the second person singular imperative form *fica* of the verb *ficar* "stay", can be interpreted as the phonetic realisation of a derived long vowel a at the phrasal level. A similar phenomenon is also observed within the domain of the prosodic word:

(83) *Dava-a*. /dava#a/ → ['da:v^a] "He/she/you used to give it."

(84) *Dava-o*. /dava#u/ → ['da:v^o] "He/she/you used to give it."

Example sentence (80) is of the same type as *Fic*[a]*qui*, and appears to derive as follows: *Dava^μ-a^μ* → *Dava^μ-a^μ* → *Dava^{μμ}*. In this process, the vowel quality of the second segment *a* is deleted by the OCP, the remaining mora is relinked to the preceding *a*, and a derived long vowel a is formed, which is realised as [a] at the surface level.

Example (81) is slightly more complex: the [−round] and [+low, +back] features of the vowel /a/ are delinked from the Labial and Dorsal nodes, respectively, in the feature geometry, allowing

the spreading of the [+round] and [-low, +back] features of the following /u/. Then, /u/ itself is delinked from its mora at the root node level, and a segment with altered feature values is relinked to the vacated mora, forming the derived long vowel $\overset{\mu\mu}{\text{ɔ}}$, as shown in (82), where the features retained from the two original segments in this coalescence are shaded. This vowel is subsequently shortened by one mora under constraint hierarchy (64), surfacing as the optimal output [ɔ].

(85)

	$\overset{\mu}{/a/}$	$\overset{\mu}{/u/}$		$\overset{\mu\mu}{\text{ɔ}}$
low	+	–	→	–
high	–	+		–
ATR	–	+		–
back	+	+		+
round	–	+		+

It is assumed here that in this spreading, the feature [+back] is delinked from /a/, because in the assimilation pattern /ai/ → [ɛ(:)], as common as /au/ → [ɔ(:)], /a/ delinks its [+back] feature and accepts the spreading of [–back] from /i/. Moreover, the changes /au/ → [ɔ(:)] and /ai/ → [ɛ(:)] exhibit a symmetrical pattern.

6.4. The Distinctive Opposition [á:] ≈ [é:] in *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives

In stressed CV syllables preceding +nasal consonants [m n ɲ], /a/ is realised as [é:], reflecting a change from [+low] to [–low] — e.g. *cama* "bed" /kama/ → ['kə:mə]; *ano* "year" /ano/ → ['e:nu]; *banho* "bath" /bapo/ or /ban'o/ → ['bɐ:ɲu]. This raising may be regarded as a type of weakening, possibly resulting from airflow redistribution: nasal resonance in vowels, triggered by the spreading of the [+nasal] feature, diverts part of the pulmonic airstream into the nasal cavity, thereby reducing the amount of airflow available for oral articulation and rendering the [+low] region inaccessible.

This phonological context also allows, nevertheless, for occurrences of the [+low] vowel [á:], though in highly restricted environments. These include the noun *g[á:]nho* "gain, profit" and the rizotonic forms of the verb *g[a]nhar* "to gain" — namely, *g[á:]nho*, *g[á:]nhas*, *g[á:]nha*, *g[á:]nham*, *g[á:]nhe*, *g[á:]nhes*, *g[á:]nhem* — as well as the first person plural of the preterite indicative in *-ar* regular verbs, encoded by the ending *-ámos* [–á:muʃ]. This final form stands in systematic contrast with the first person plural of present indicative ending *-amos* [–é:muʃ], as

exemplified by *fal[á:]mos* "we spoke" vs. *fal[é:]mos* "we speak", both derived from the verb *falar* "to speak".

Accordingly, even in the context preceding [+nasal] consonantal segments, the mid vowel [é:] and the low vowel [á:] stand in a distinctive opposition. The vowel *a* in the root *ganh-* is historically a long vowel of bimoraic weight, as noted by Teyssier (1981: 42): "Temos, com efeito, *ga-anha* > *ganha* (verbo) e *ga-anho* > *ganho* (substantivo), nos quais o *a*, resultante da contracção, conservou até hoje no português europeu um timbre aberto [a], apesar da presença da consoante nasal seguinte, que, nas palavras que contêm um *a* singelo etimológico, sempre fechou esta vogal em [ä]" — that is, [ɐ] or [ə] in IPA. And given that it does not undergo quality reduction even in unstressed syllables, as in *g[a]nhar*, it is reasonable to assume that it still retains its bimoraic status in contemporary usage, reflecting the alternation schema $\overset{\mu\mu}{/a/} \rightarrow [\acute{a}:] \approx [a]$ discussed in (15). This vowel length may thus be interpreted as blocking the weakening process that would otherwise occur before nasal consonants.

As previously noted in Makino (2007¹: 3), "in the preterite indicative, each combination of 'indicative-nonpast-finished' and a specific 'person-number' set appears morphologically as an *amalgam* — e.g. *fale-i*, *fala-ste*, *falo-u*, *falá-mos* (← *fala-* + $\overset{\mu}{-}^{*}\text{mos}$), *fala-stes*, *fala-ram*" (translated from Japanese). On the notion of *amalgam* — *amalgame* in French — Martinet (1991: 102) writes: "deux signifiés qui coexistent dans un énoncé enchevêtrant leurs signifiants de telle façon qu'on ne saurait analyser le résultat en segments successifs". The first person plural ending of the preterite indicative $\overset{\mu}{-}^{*}\text{mos}$ — that is, $\overset{\mu}{-}^{*}\text{mos}$ in the example above — is thus a morphological amalgam of [–non-assertive, –VP∈past, –finished, –assumptive] with [–non-participant, –addressee, +plural]. It **begins not with a segmental element but with a floating mora** $\overset{\mu}{/}$, much like "the Hebrew definite prefix *ha*", which "has the phonological form *haX*, where *X* is an empty position that prefers to be occupied by the following consonant but when this is impossible then by the preceding vowel" (Kenstowicz 1994: 44–45). In contrast, the *-mos* of the present indicative encodes only the person-number features [–non-participant, –addressee, +plural] and therefore lacks such a floating mora $\overset{\mu}{/}$. If one assumes then that the theme vowel *a* of the verbal stem — for example, the vowel *a* of the stem *fala-* of the verb *falar* — becomes linked to this initial mora, yielding a bimoraic structure as in (83), the form *fal[á:]mos* should be analysed on a par with $\overset{\mu\mu}{/ga\eta o/} \rightarrow g[\acute{a}:]nho$. By contrast, in (84), such mora linking does not occur due to the absence of a floating mora:

(86) preterite indicative: $\overset{\mu}{fala-} + \overset{\mu}{-}^{*}\text{mos} \rightarrow \overset{\mu}{fala-} \overset{\mu}{mos} \rightarrow \overset{\mu\mu}{fala-mos} \rightarrow fal[\acute{a}:]mos$

(87) present indicative: $fala^- + -mos \rightarrow fal\bar{a}^{-}mos \rightarrow fal[\acute{e}:]mos$

As discussed in section 6.1, the tautosyllabic vowel-nasal sequence Vn undergoes nasalisation of V and deletion of /n/ — in en ɛn an ɔn on un → ɪ̃: ẽ: ẽ: ẽ: õ: õ: û: — prior to the Vowel Quality Modification Tier, and the tongue height raising of an → ẽ: during this process can be attributed to the influence of [+nasal] from the nasal segment n. Similarly, as argued just above, the raising of short á to é in stressed CV syllables preceding [m, n, ɲ] is likewise attributable to the [+nasal] feature from these consonants. Given that both the raising in a[+nasal]. → ẽ: and the raising in á.[+nasal] → é.[+nasal] — where . denotes a syllable boundary — are to be triggered by the same feature [+nasal], it seems unlikely that they should occur at different tiers. If we therefore assume that the latter process also precedes the Vowel Quality Modification Tier, then the short vowel /a/ and the long vowel /a/ that occur in stressed CV syllables before [+nasal] consonants would be realised as é and á:, respectively, at the input to that tier. At this stage, they are evaluated as optimal intermediate outputs by the set of constraints (58) operative in this stratum, and passed on as inputs to the subsequent Vowel Quantity Adjustment Tier. As shown in the tableaux in (85) and (86), they are then output as [é:] and [á:], respectively:

(88) $falamos /fal-a^{-}mos/$

$fe'le\mu f$	ID-QUAL	MAX	DEP	* $\mu\mu\mu$	*' $\sigma[\mu]$	* $\mu\mu/V$	MAX(μ)	*' $\sigma[\mu]$	DEP(μ)
$\text{☞ } fe'le:\mu f$						*			*
$fe'le\mu f$					*W	L		*W	L
$fe'lamu f$	*W				*W	L		*W	L
$fe'le:\mu f$			*W						*

(89) $falámos /fal-a^{-}mos/$

$fe'la:\mu f$	ID-QUAL	MAX	DEP	* $\mu\mu\mu$	*' $\sigma[\mu]$	* $\mu\mu/V$	MAX(μ)	*' $\sigma[\mu]$	DEP(μ)
$\text{☞ } fe'la:\mu f$						*			
$fe'lamu f$					*W	L	*W	*W	
$fe'le:\mu f$	*W					*			*

In the preterite indicative of regular *-ar* verbs, the endings *-i*, *-ste*, *-u*, *-stes*, and *-ram* are all morphological amalgams encoding both the TAM feature bundle [–non-assertive, –VP∈past, –finished, –assumptive] and the person–number values [±non-participant, ±addressee, ±plural]. In contrast, the present indicative endings — *-o*, *-s*, *∅*, *-is*, and *-m* — encode only the person–number values [±non-participant, ±addressee, ±plural] and lack any temporal-aspectual

specification. It is therefore highly implausible, from a systemic point of view, to assume that only the first person plural of the preterite should share its ending *-mos* with the present indicative. Such an assumption would disrupt the overall morphological and semantic parallelism of the system and thus seems theoretically unnatural.

Accordingly, the surface contrast between [é:] and [á:] in stressed CV syllables preceding nasal consonants [m n ɲ] is best interpreted as the phonetic realisation of an derived opposition between $\overset{\mu}{a}$ and $\overset{\mu\mu}{a}$ — that is, between a monomoraic short vowel and a bimoraic long vowel at Lexical Stratum 1: Feature Establishment in (18). On this analysis, thus, stressed CV syllables preceding [+nasal] consonants [m n ɲ] constitute the only type of stressed syllable in which the opposition between the short vowel $\overset{\mu}{a}$ and the long vowel $\overset{\mu\mu}{a}$ does not undergo neutralisation — precisely because of the quality weakening affecting the short vowel.

On the other hand, the situation is different for regular verbs of the *-er* and *-ir* conjugations. The theme vowels *e* and *i* of these verbs are the underlying short vowels $\overset{\mu}{e}/$ and $\overset{\mu}{i}/$, respectively. When these vowels are linked to the floating mora $\overset{\mu}{/}$ of the preterite indicative first person plural ending *-mos*, they form derived long vowels $\overset{\mu\mu}{e}$ and $\overset{\mu\mu}{i}$. In stressed CV syllables, however, the surface realisation of these derived long vowels is identical to that of their short counterparts — both are realised phonetically as [é:] and [í:], respectively, and thus the opposition is neutralised at the phonetic level. This interpretation is based on the following inference: as shown in (17), the underlying long vowels $\overset{\mu\mu}{/e}/$, $\overset{\mu\mu}{/a}/$, and $\overset{\mu\mu}{/o}/$ are mapped to [é:], [á:], and [ó:], respectively, in stressed CV syllables and, If systemic consistency and parallelism are to be maintained, then the derived long vowels $\overset{\mu\mu}{e}$ and $\overset{\mu\mu}{i}$ must likewise be mapped to [é:] and [í:], which coincide with the mapping of the underlying short vowels $\overset{\mu}{e}/$ and $\overset{\mu}{i}/$ in the same prosodic environment, also illustrated in (17). This is illustrated in (87) and (88) by comparing the first person plural form *comemos* in the preterite indicative with the superficially identical form *comemos* in the present indicative of the verb *comer* "to eat".

(90) preterite indicative: $\overset{\mu}{come}- + \overset{\mu}{-mos} \rightarrow \overset{\mu}{come}-\overset{\mu}{mos} \rightarrow \overset{\mu\mu}{come}-mos \rightarrow com[\overset{\mu\mu}{e}:]mos$

(91) present indicative: $\overset{\mu}{come}- + -mos \rightarrow \overset{\mu}{come}-mos \rightarrow com[\overset{\mu}{e}:]mos$

6.5. Cases Requiring Further Consideration

(i) Semantic Transparency of Morphemes and their Prosodic Structure

In some cases, semantic transparency of morphemes and their prosodic structure appear to be

related to whether a vowel undergoes reduction. In *televisão* “television” and *telefone* “telephone”, the prefix *tele-* /tɛlɛ-/ meaning “remote” is realised with reduced vowels as [tɪlɪ-]. By contrast, in *telemóvel* “mobile phone”, the same string /tɛlɛ-/ originates as a clipped form of *telefone*, and is realised with non-reduced vowels [tɛlɛ-]. While this semantic difference may play a role in the preservation of full vowel quality, it is also possible that, in *telemóvel*, the element *tele-* constitutes a prosodic word on its own, as a semantically transparent lexical constituent — unlike the *tele-* in *televisão* or *telefone*. In that case, the primary stress of the prosodic word would fall on [ˈtɛ-]. The remaining issue concerns the final unstressed vowel [-ɛ], but this may be treated analogously to the final unstressed [-ɔ] in *euro*, when pronounced as [ˈeʊrɔ].

In Makino (2010), based on the observation that native speakers perceive the stress position in *papel* and *doutor* to be unmarked — just like the stress position in *futuro*, *camada*, and *solene* — a theme vowel consisting solely of one mora without segmental quality, namely *-0* — treated as having the same prosodic status as the theme vowels *-o*, *-a*, *-e* in *futur-o*, *camad-a*, *solene-e*, referred to in Portuguese linguistics as *índices temáticos* — was postulated, and *papel* and *doutor* were thereby allowed to be analysed as *papel-0* and *doutor-0*. On this basis, all of these words were assumed to receive their primary stress through binary foot parsing as in fu(ˈtu.r-0), ca(ˈma.d-a), so(ˈle.n-e), pa(ˈpe.l-0), dou(ˈto.r-0). In contrast, words such as *câmar-a* and *túnel-0* were assumed to contain the morphs *camar-* and *tunel-*, which behave as lexical exceptions in stress assignment by requiring that their final syllable initiate binary foot parsing when adjacent to an *índice temático* *-o*, *-a*, *-e*, or *-0*. Consequently, stress is assigned through prosodification as (ˈcâ.ma)r-a and (ˈtú.ne)l-0.

Accordingly, since *tele-* and *euro* each constitute a single morpheme — with neither **tel-* nor **eur-* bearing autonomous meaning — the prosodic word structures of *tele-* and *euro* are to be understood as (ˈcâ.ma)r-a and (ˈtú.ne)l-0, respectively, and are thus prosodified as (ˈte.le)-0 and (ˈeu.ro)-0. The corresponding underlying representations are /tɛlɛ-/ and /eʊrɔ-/, in which the radical final vowels /ɛ/ and /ɔ/ become associated with the floating mora of the *índice temático* *-0* to form derived long vowels $\overset{\mu\mu}{\epsilon}$ and $\overset{\mu\mu}{\omicron}$. The surface vowels [ɛ] and [ɔ] of *tele-* and *euro* thus appear to reflect these derived long vowels.

(ii) Derived nouns and adjectives formed by suffixation directly to the verbal radical without an intervening thematic vowel.

In Portuguese, one common method of deriving nouns and adjectives from verbs involves the addition of suffixes such as *-ção*, *-dor*, *-tiv-o*, and *-tóri-o* to the verbal radical. In this derivational process, a theme (or thematic) vowel is typically inserted between the verbal radical and the suffix (e.g. [*compar-a*]*r* → [*compar-a*]*ção*; [*bat-e*]*r* → [*bat-e*]*dor*; [*part-i*]*r* → [*part-i*]*tiv-o*; [*fal-a*]*r* → [*fal-a*]*tóri-o* — square brackets indicate the verbal stem, and underlining marks the verbal radical). In these derived nouns and adjectives, word stress falls on the suffix — *-ção*, *-dor*, *-tiv-*, *-tóri-* — rendering the radical vowel unstressed. When this vowel is *a*, it undergoes quality reduction — e.g. [*comp*][*ɐ*]*r-a*ção; [*b*][*ɐ*]*t-e*dor; [*p*][*ɐ*]*rt-i*tiv-o; [*f*][*ɐ*]*l-a*tóri-o.

In contrast to this type of word formation, when such suffixes are attached directly to the verbal base without an intervening thematic vowel, they take the forms *-ão* (underlyingly *-ion*), *-or*, *-iv-o*, *-óri-o*, and the radical vowel *a* often does **not** undergo reduction as in (92):

$$(92) [dil][\mathbf{v}]t\text{-}\mathbf{a}r \rightarrow [dil][\mathbf{v}]t\text{-}\mathbf{a}ção, [dil][\mathbf{v}]t\text{-}\mathbf{a}dor \Leftrightarrow [dil][\mathbf{a}]\mathbf{\check{c}}ão, [dil][\mathbf{a}]tóri-o$$

$$[per\text{-}su[\mathbf{v}]d\text{-}\mathbf{i}]r \rightarrow [per\text{-}su[\mathbf{v}]d\text{-}\mathbf{i}]ção \Leftrightarrow [per\text{-}su[\mathbf{a}]s]ão, [per\text{-}su[\mathbf{a}]s]iv-o, [per\text{-}su[\mathbf{a}]s]óri-o$$

$$[in\text{-}f[\mathbf{i}]r\text{-}\mathbf{i}]r \rightarrow [i\text{-}l[\mathbf{a}]\mathbf{\check{c}}]ão, [i\text{-}l[\mathbf{a}]t]iv-o$$

The last example involves a suppletive root alternation between *fer-* and *lat-*, but is included here because the lack of vowel reduction appears to correlate with the morphological structure of the word. Barbosa (1988: 353) makes an observation of particular interest in this regard, noting that "... the use of [a] has extended to words like *dissuasão* and *persuasão*, where it alternates with the standard [ɐ]...". This suggests that **the non-reduction of the root vowel *a* in the morphological context [verbal base + non-thematic vowel] + [suffix *-ão*, *-iv-*, *-or*, *-tóri-*]** is a phenomenon currently in progress from a diachronic perspective.

It remains unclear whether this phenomenon results from analogy with some existing paradigm. From a morphophonological standpoint, however, it is possible to account for the data by assuming that the stem vowel *-a*, even when not reduced, is underlyingly a short vowel /*a*^μ/. On this basis, in surface forms that appear to lack a stem vowel, one may posit, as shown in (93), a stem vowel *-0* with no segmental quality but with a vocalic weight of one mora — just like the other stem vowels *-a*, *-e*, *-i* — thereby rendering the phenomenon theoretically tractable:

$$(93) [dilat\text{-}\mathbf{a}^{\mu}]r \rightarrow [dilat\text{-}\mathbf{a}^{\mu}]\mathbf{\check{c}}ão, [dilat\text{-}\mathbf{a}^{\mu}]dor$$

$$[dilat\text{-}\mathbf{a}^{\mu}]r \rightarrow [dila\mathbf{\check{c}}\text{-}\mathbf{0}^{\mu}]ão, [dilat\text{-}\mathbf{0}^{\mu}]óri-o$$

Then, in forms with the stem vowel *-0*, we may assume that the radical vowel /*a*/ is relinked to

the floating mora once associated with *-o*, yielding a derived long vowel /ḁ̥̥/. This /ḁ̥̥/, like an underlying long vowel /ḁ̥̥/, resists quality reduction even in unstressed syllables as shown in (94):

$$(94) \text{dila}^{\mu}\text{a}^{\mu}\text{ç}^{\mu} - \tilde{a}o \rightarrow \text{dila}^{\mu\mu}\text{a}^{\mu\mu}\text{ç}^{\mu\mu} - \tilde{a}o \rightarrow \text{dil}[\text{a}]^{\mu\mu}\text{ç}^{\mu\mu} - \tilde{a}o$$

The validity of this interpretation remains open to further investigation, but it seems clear that forms in which the stem vowel *a* resists reduction tend to lack a thematic vowel, and that this absence is closely related to the lack of reduction.

(iii) Metrical Foot Structure and Vowel Reduction

The present study, as outlined above, has accounted for vowel reduction and non-reduction by means of the underlying or derived mora count of vowels, the presence or absence of primary stress in syllables headed by those vowels, and the theoretical apparatus of constraint hierarchies operating within the Vowel Quality Modification Tier and the Vowel Quantity Adjustment Tier. However, there still remain phenomena that cannot be explained by these factors alone. For instance, *ecónom-o* “manager; steward”, *económ-ico* “economic”, and *econom-ia* “economy”, which all share the word constituent *econom-*, exhibit the following vowel alternations, as shown in (95):

$$(95) \text{ec}[\acute{o}]n[\text{u}]m-o \approx \text{ec}[\text{u}]n[\acute{o}]m-ico \approx \text{ec}[\text{ɔ}]n[\text{u}]m-ia$$

Judging from the alternation $\text{ec}[\acute{o}]n[\text{u}]m-o \approx \text{ec}[\text{u}]n[\acute{o}]m-ico$, the underlying form of the radical in these words appears to be /iḳoṇoṃ-/, where the first /ɔ̥̥̥/ is, like the second /ɔ̥̥̥/, an underlying short vowel. In $\text{ec}[\text{ɔ}]n[\text{u}]m-ia$, nevertheless, this underlying short vowel /ɔ̥̥̥/ is realised as [ɔ], showing no reduction. One possible explanation lies in the difference in prosodic structure, as illustrated in (96):

$$(96) (,i.ḳ\acute{o})('n\acute{o}.mi)ko \approx i(,ḳ\acute{o}.n\acute{o})('mi.a)$$

In *económico*, where the underlying /ɔ̥̥̥/ is realised as reduced [u], the vowel occurs in the complement syllable in the foot (,i.ḳ\acute{o}). In contrast, in *economia*, where the same underlying /ɔ̥̥̥/ is realised as unreduced [ɔ], it occurs in the head syllable of the foot (,ḳ\acute{o}.n\acute{o}). This prosodic difference appears to be related to whether vowel reduction takes place, suggesting that further investigation and research on this point is required.

(iv) Derived words with root vowels in hiatus with a preceding vowel

The root vowels $\overset{\mu}{/e/}$ and $\overset{\mu}{/ɛ/}$, when placed in vowel sequences such as *-ae-*, *-ie-*, *-oe-*, *-ue-* — e.g. in *baetilha*, *ajaezar*, *alienar*, *enviesar*, *admoestar*, *amoedar*, *duetista*, and *duelista* — generally do not undergo reduction in contemporary EP, even when they do not bear stress in derivation or verbal inflection, as shown in (97):

(97) $b[\text{v}'e:]t-a \approx b[\text{v}.e]t-ilh-a$; $j[\text{v}'e:]z \approx a-j[\text{v}.e]z-a-r$; $al[i'e:]n-a \approx al[i.e]n-a-r$;

$v[i'e:]s \approx en-v[i.ɛ]s-a-r$; $adm[u'ɛ:]st-a \approx adm[u.ɛ]st-a-r$; $m[u'ɛ:]d-a \approx a-m[u.ɛ]d-a-r$;

$d[u'e:]t-o \approx d[u.e]t-ist-a$, $d[u.e]t-íst-ic-o$; $d[u'ɛ:]l-o \approx d[u.ɛ]l-ist-a$

On the other hand, in Fonseca 1957, many of these examples are transcribed with reduced vowels such as [i] or [ɪ] — e.g. $a-j[\text{v}.i]z-ar$, $b[\text{v}.i]t-ilha$, $adm[u.i]st-ar$, $a-m[u.i]d-ar$, $d[u.i]l-ita$ — suggesting that the phenomenon whereby unstressed vowels remain unreduced in vowel hiatus contexts is likely to be in progress diachronically. Furthermore, the vowel of the inchoative suffixes *-ec-* and *-esc-* consistently shows reduction after more open vowels in all sources — e.g. $esm[\text{v}'e:]ce \approx esm[\text{v}.i]cer$, $esv[\text{v}'e:]ce \approx esv[\text{v}.i]cer$ — but not after more closed vowels, as in $aqu[i'ɛ:]sce \approx aqu[i.ɛ]scer$. This suggests that it is not hiatus alone that determines whether a vowel undergoes reduction, but also whether it belongs to a root or an affix, as well as the aperture relation between the preceding and following vowels. This is therefore a case that calls for further detailed investigation.

7. Conclusion

Although the concept of mora has not traditionally been applied to the analysis of Portuguese vowels, the present study adopts this notion and posits an opposition between the underlying short vowels $\overset{\mu}{/i} \overset{\mu}{/e} \overset{\mu}{/ɛ} \overset{\mu}{/a} \overset{\mu}{/ɔ} \overset{\mu}{/u/}$ and the underlying long vowels $\overset{\mu\mu}{/ɛ} \overset{\mu\mu}{/a} \overset{\mu\mu}{/ɔ/}$ in Contemporary European Portuguese. Furthermore, by establishing a system of vowel quality reduction and vowel quantity adjustment arising from the constraint hierarchies operating in two postlexical strata — Postlexical Stratum 1 and Postlexical Stratum 2 — we attempted to account for two types of vowel alternation — $[í: \acute{e}: \acute{e}: \acute{a}: \acute{o}: \acute{u}:] \approx [i \text{ } i \text{ } i \text{ } v \text{ } u \text{ } u]$ and $[\acute{e}: \acute{a}: \acute{o}:] \approx [\varepsilon \text{ } a \text{ } ɔ]$ — observed at the phonetic level. Building on this descriptive account, we showed that it is also possible to provide a unified explanation for a range of apparently unrelated phonetic phenomena: the

absence of vowel quality reduction in nasal vowels in unstressed syllables; the characteristic vowel alternations in open syllables preceding prepalatal and palatal consonants [ʃ ʒ ʎ ɲ]; the phonetic realisation of vowel coalescence at the level of phrasal phonology; the opposition [é:] ≈ [á:] in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives; vowel opening in radical vowels found in derived nouns and adjectives formed by suffixation to verbal radicals lacking intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative implicatures.

Finally, in traditional accounts — whether explicitly or implicitly — it has been assumed that whether an unstressed vowel in a given word undergoes reduction or not is synchronically unpredictable, and must therefore be memorised individually at the lexical level. In other words, speakers are assumed to store not only the "underlying segmental sequence" of each word, but also an additional piece of information regarding whether vowel reduction applies or not. This results in a high degree of lexical redundancy. By contrast, under the present analysis, it suffices in most cases for the speaker to memorise only the underlying segmental sequence of a word at the lexical level. The presence or absence of vowel reduction becomes predictable from the evaluation of output candidates by the constraint rankings operating in each of the lexical and postlexical strata of phonological derivation. As a result, in most cases, the only information that must be stored is the phoneme sequence along with vowel length. This yields a representation that is more compact and principled than in traditional theories, which rely on rote memorisation of exceptional reduction behaviour. At the same time, however, the existence of alternation patterns such as *ec[ɔ:]n[u]m-o* ≈ *ec[u]n[ɔ:]m-ico* ≈ *ec[ɔ]n[u]m-ia* highlights the need for further investigation and research, as well as for refinement of the theoretical framework.

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